

**BIOSPHERE: A BIOLOGICAL MUSEUM MANAGEMENT SYSTEM FOR
SPECIMEN INVENTORY AND VISITOR SERVICES**

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ABSTRACT

The University of San Carlos Biological Museum currently relies on paper-based records, manual procedures, and spreadsheets to document and manage its biological specimens. These fragmented practices make specimen retrieval, record updating, storage-location monitoring, report generation, and long-term data preservation more difficult as the collection continues to grow. The absence of a centralized backup and recovery mechanism also raises concerns regarding the accidental deletion, loss, or inaccessibility of important specimen records. This study proposes BioSphere, an offline-first web-based system designed to centralize specimen inventory management and support selected museum visitor and exhibit services. The system will provide authenticated curator access for managing specimen records, cataloging, Uncataloged records, storage locations, reports, visit requests, general inquiries, and QR-based exhibit content. It will also include a public museum website, a Controlled Knowledge-Based FAQ Assistant, unlisted QR exhibit pages, and selected web-based augmented reality experiences while restricting public access to internal collection and storage information. Limited developer functions will support authorized curator-account administration and the deployment of approved AR models. Offline capability will be provided as a secondary feature limited to viewing and searching previously synchronized specimen records and creating new text-only Uncataloged records that can be submitted after reconnection. The project will progress sequentially through the phases of requirements analysis, system design, implementation, testing, deployment, and maintenance. BioSphere is expected to improve the accessibility, availability, organization, and preservation of museum records while reducing dependence on fragmented documentation. The completed system will be evaluated through usability, reliability, security, and user acceptance testing to determine whether it satisfies the museum's validated requirements.

Keywords: biological specimen inventory, museum information system, QR-based exhibits, augmented reality, offline-first web system

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CHAPTER 1

INTRODUCTION

This chapter presents the foundation of the study by discussing its rationale, statement of the problem, specific objectives, significance, and scope and limitations. These sections provide the context, purpose, and direction of the research while defining its coverage and intended contributions.

1.1 Rationale of the Study

The University of San Carlos Biological Museum (University of San Carlos, n.d.) serves as an academic and educational resource that supports biological instruction, research, collection preservation, and public appreciation of biodiversity. As part of the Department of Biology, the museum holds biological specimens accumulated through years of academic work, field collection, preservation, and curatorial care. These specimens represent valuable scientific and educational materials that must be properly documented, organized, preserved, and made accessible to authorized users and intended visitors.

At present, the museum primarily relies on manual records, paper-based files, and spreadsheets to document and manage its biological specimens. While this approach has supported daily operations, it becomes increasingly inefficient as the collection grows, making specimen retrieval, record updates, identification of possible duplicate records, report generation, and long-term data preservation more challenging. In addition, the museum has limited digital support for visitor coordination and public information, as inquiries and visit requests are handled through manual communication channels.

Existing museum management systems such as Symbiota, Arctos, Specify, and PastPerfect Museum Software demonstrate the value of centralized specimen cataloging, record retrieval, location tracking, and report generation (Gries et al., 2014; Cicero et al., 2024; Specify Collections Consortium, 2018; PastPerfect Museum Software, 2024). Augmented reality platforms such as

AssemblrWorld and Animal Safari AR also demonstrate how selected digital experiences can enhance visitor engagement. These systems provide relevant design references for integrating centralized inventory management, QR-based exhibits, selected Augmented Reality (AR) features, and a Controlled Knowledge-Based Frequently-Asked-Questions (FAQ) Assistant that answers only from curator-approved museum information and redirects unanswered questions to the General Inquiry form.

In response to these needs, this study proposes BioSphere: An Offline-First Biological Museum Management System with Quick Response (QR) based Exhibits and AR, a centralized platform designed to improve specimen inventory management, visitor coordination, and public access to approved museum information. The system will centralize specimen and storage-location records, support data import and possible duplicate-record warnings, process general inquiries and visit requests, manage QR-based exhibit content, provide selected augmented reality experiences, and offer a Controlled Knowledge-Based FAQ Assistant. The curator application will provide limited offline continuity by allowing previously synchronized specimen records to be viewed and searched and by allowing new Uncataloged text-only drafts to be recorded during temporary connectivity interruptions. These drafts will be submitted after connectivity is restored. Through these capabilities, BioSphere aims to reduce fragmented documentation, improve record accessibility, strengthen the museum's digital presence, and provide a practical, secure, and sustainable solution for the USC Biological Museum.

1.2 Statement of the Problem

1.2.1 General Objective

This study aims to design and develop BioSphere: An Offline-First Biological Museum Management System with QR-Based Exhibits and Augmented Reality to improve specimen management, visitor

coordination, and access to approved museum information at the University of San Carlos Biological Museum.

1.2.2 Specific Objectives

Specifically, the study aims to:

1. Identify and analyze the museum's current specimen, storage, public information, inquiry, and visit-request workflows and requirements.
2. Design and develop BioSphere as an offline-first web-based biological museum management system.
3. Test and evaluate the system in terms of functional suitability, usability, reliability, performance efficiency, security, and maintainability.
4. Deploy BioSphere to University of San Carlos Biological Museum.

1.3 Significance of the Study

These are the entities that will benefit this study:

Curator

It will reduce administrative workload by providing tools for managing specimen records, storage locations, manual encoding and spreadsheet imports, possible duplicate-record warnings, curator account succession, public exhibit visibility, QR exhibit content, selected AR deployment references, visitor requests, general inquiries, reports, and visitor information summaries for submission through the University's external campus-entry process.

Visitors

It will improve the museum experience by providing convenient access to approved museum information, a rule-based Frequently-Asked-Questions assistant, QR-based exhibit content, selected augmented reality experiences, visitor guidelines, general inquiry forms, and an organized visit-request process.

Students and Faculty

It will provide students and faculty with easier access to curator-approved specimen information, QR-based exhibit resources, and selected augmented reality experiences that may support classroom instruction, academic activities, and learning.

University

It will strengthen the University's digital transformation initiatives by modernizing museum operations, improving specimen management, and enhancing public engagement through an accessible online platform.

Academic Institutions

It may serve as a reference for institutions seeking to adopt digital technologies for specimen inventory management, museum administration, and interactive educational exhibits.

Museums and Collection-Holding Institutions

It may demonstrate how centralized inventory management, QR technologies, offline-first web functions, and selected augmented reality experiences can be adapted by small museums, university museums, and academic collections with similar operational and resource constraints.

Researchers

It may support authorized researchers by enabling the curator to retrieve, organize, and provide approved specimen information more efficiently for biodiversity studies, taxonomic research, ecological investigations, and other

scientific and academic activities. Access to confidential inventory records and internal storage information will remain subject to museum authorization.

Future Researchers

It may serve as a reference and foundation for future studies involving biological museum management systems, offline-first web applications, augmented reality, QR-based exhibits, and related digital technologies for museums.

1.4 Scope and Limitations

This study covers the analysis, design, development, pilot deployment, and evaluation of BioSphere, an offline-first web-based biological museum management system for the University of San Carlos Biological Museum (USC-BM).

The authenticated curator application will support account management, specimen cataloging, storage-location management, manual encoding, spreadsheet and Comma-Separated Values (CSV) import, possible duplicate-record warnings, inquiry and visit-request management, reporting, data export, QR exhibit management, audit logging, and backup and recovery. Authorized curators may create, activate, or deactivate curator accounts to support account succession and handover. Approved curator onboarding will use a secure email invitation for password setup, while forgotten-password recovery will use a secure, time-limited email link. Authenticated Curator and Developer accounts may change their own passwords through BioSphere.

Developer access will be restricted to onboarding the initial curator, managing curator access when formally authorized, and deploying or updating approved AR assets. These functions will be performed through a restricted developer interface, with controlled backend procedures used only when necessary. Initial curator onboarding will trigger a secure password-setup invitation rather than requiring the Developer to assign or transmit a password.

The public website will provide approved museum information, visitor guidelines, a Controlled Knowledge-Based FAQ Assistant, General Inquiry and Visit Request forms, email notifications to the authorized curator for new submissions, curator-initiated outbound inquiry replies and Visit Request updates to visitors, unlisted QR exhibit pages, and selected AR experiences. BioSphere will not access or synchronize an external email mailbox. Replies received through the museum's official email address will remain outside BioSphere and may be manually reflected in the corresponding record by the curator. The Home, Gallery, Visit, and About page information will be implemented as approved static frontend content. BioSphere will not provide the curator with a general content-management interface for editing these pages. Subsequent changes will require an approved source-code update and redeployment by an authorized developer or designated USC/DCISM maintainer. This limitation does not apply to curator-managed FAQ knowledge or QR exhibit pages.

Offline support will be limited to viewing and searching previously synchronized specimen records and creating new Uncataloged text-only drafts that are submitted after reconnection. The pilot implementation will use approximately 500 curator-provided and validated specimen records from the museum's main gallery, subject to availability and client approval. BioSphere will be evaluated in terms of functional suitability, usability, reliability, performance efficiency, security, maintainability, limited offline record access, offline-draft preservation and synchronization, and user acceptance.

The study is limited to the development and evaluation of BioSphere for USC-BM. Implementation in other museums will require further customization. BioSphere will not replace the University's official campus-entry procedures or other university-managed systems.

Public users will have access only to curator-approved museum information. Full specimen records, detailed storage locations, curator accounts, audit logs, and operational data will remain restricted to authorized users. The

public website will not provide a searchable specimen catalog. QR exhibit pages will remain unlisted, although users with a valid link may access their approved public content.

Offline-first functionality will apply only to read-only access to previously synchronized specimen records, local search and filtering, and creation of new Uncataloged text-only drafts. Existing specimen records, storage assignments, lot quantities, condition values, images, imports, archives, reports, General Inquiries, Visit Requests, account functions, public-content publishing, email notifications, QR publishing, backup and recovery, and AR experiences will require internet connectivity. Offline-created drafts will remain stored in the authorized browser until synchronization succeeds or the curator intentionally discards them. The MVP will not include offline editing of existing records or an online-versus-offline conflict-resolution interface.

The Controlled Knowledge-Based FAQ Assistant will answer only from curator-approved museum information. When no approved answer is available, it will direct the visitor to the General Inquiry form. It will not identify species, generate scientific conclusions, process or approve visit requests, or replace communication with museum personnel.

Spreadsheet and CSV imports will follow an approved BioSphere template with defined fields and column mappings. Existing files may require restructuring, cleaning, or validation before import. Paper records must be manually encoded by authorized museum personnel. Full digitization and migration of the museum's historical collection are outside the team's responsibility.

Possible duplicate-record warnings will be based on selected matching fields and will not automatically confirm that records are duplicates. Final review and correction will remain the curator's responsibility.

The initial AR implementation will target approximately 20 curator-selected specimens with suitable 3D assets. The final number may change depending on

modeling difficulty, development time, asset quality, device compatibility, and testing results. QR exhibit labels will provide both a scannable QR code and a human-readable direct URL to the corresponding exhibit page so that visitors whose devices cannot scan the QR code may still access the approved digital content through a web browser. AR will function as an optional enhancement for compatible devices rather than a requirement for accessing exhibit information. When AR is unsupported, unavailable, denied, or fails to load, visitors will retain access to the regular exhibit page containing approved images and textual information and, where technically supported, the standard three-dimensional model viewer.

Specimens for augmented reality presentation will be selected under the curator's discretion based on their relevance to a particular research or educational objective. Selection may consider conservation status, rarity, endemism, exotic or invasive status, and economic or ecological value. These factors serve as selection guidelines rather than a fixed scoring system, as their relevance may vary depending on the specimen group and intended objective.

Backup and recovery will cover only records and supported files stored within BioSphere's approved database and file-storage environment. The system will not recover paper records, unsupported external files, or information that was never entered into the system.

BioSphere will exclude online payments, ticket sales, automatic campus-entry approval, AI-based species identification, laboratory management, financial management, and unrestricted public access to specimen records. Long-term hosting, maintenance, additional AR development, and institutional adoption beyond the capstone period will remain the responsibility of USC-BM and the appropriate University units.

CHAPTER 2

RELATED SYSTEMS

This chapter reviews existing systems and supporting studies relevant to the development of BioSphere. The discussion is organized according to the operational needs identified in Chapter 1: biological specimen documentation and retrieval, continuity of inventory operations during connectivity interruptions, structured visitor requests and inquiries, and QR-based exhibit information with selected augmented reality features. The reviewed systems are analyzed according to their relevant functions, limitations, and applicability to the proposed system. The chapter concludes with a comparative analysis and identifies the functional gap that BioSphere is intended to address.

Museums, and Biological Museums in Particular

The International Council of Museums defines a museum as a not-for-profit, permanent institution in the service of society that researches, collects, conserves, interprets, and exhibits tangible and intangible heritage, while remaining open, accessible, and inclusive to the public (International Council of Museums [ICOM], 2022). This definition frames a museum's core functions as simultaneously custodial and communicative: institutions are expected to safeguard the material they hold while also making it meaningful to visitors.

A biological or natural history museum applies these same functions to specimens of biodiversity, plants, animals, fungi, and other organisms collected, preserved, and catalogued for research, education, and conservation purposes. Because these specimens are physical, finite, and often irreplaceable, biological museums carry an added responsibility: their records must be accurate enough to support scientific reference and specimen recovery, while their exhibits must still communicate ecological and taxonomic information to a general audience (Abo Zeid, 2022). The USC Biological Museum operates within this same dual

role. It maintains a working specimen collection for academic and research use while also receiving visitors and inquiries from the public, which means any supporting system must serve both internal inventory needs and outward-facing engagement, rather than one at the expense of the other.

Common Problems Museums Currently Face

Inventory and collection management that is manual, offline-fragile, or incomplete. A recurring problem across the literature is that many museums still rely on manual or spreadsheet-based inventory processes that leave records incomplete, outdated, or difficult to reconcile as collections grow (Harapan & Widodo, 2021). This is not limited to small or resource-constrained institutions: even the Natural History Museum, London, which holds tens of millions of specimens, continues to face a multi-decade digitization backlog despite sustained investment, underscoring how persistent and scale-independent this problem is. Funding and staffing constraints are consistently cited as the primary barriers to digitizing and maintaining specimen records, particularly for smaller institutions operating without dedicated IT support (Ong et al., 2023). A related but distinct problem is connectivity: systems designed only for constant internet access become unusable in the exact conditions many museums and field or storage areas actually operate under, which is why offline-first architecture, rather than an occasional convenience, is treated in the literature as a core reliability requirement (Sirigiri et al., 2023).

No structured channel for visitor requests, inquiries, or curator meetings. Museums that lack a dedicated request-handling system tend to fall back on ad hoc communication, phone calls, walk-ins, or informal messages, that are difficult for administrators to track, confirm, or prioritize. The reservation-systems literature shows that adopting even a simple structured platform improves booking efficiency and reduces staff workload, but also that adoption is shaped as much by staff readiness and accessibility as by the technology itself (Ankor et al., 2024). At the same time, museums that field a high

volume of repetitive visitor questions, hours, requirements, directions, general information, without any self-service option place a continuous, avoidable load on staff time, which is the gap a lightweight FAQ Assistant is meant to close (Parvathi et al., 2024).

Static exhibits that under-communicate complex biological information. Museums that rely solely on static displays and fixed interaction styles tend to increase visitors' cognitive load and reduce engagement, particularly for content, such as ecological processes or taxonomic relationships, that is difficult to visualize from text and images alone (Spadoni et al., 2022). This leaves visitors in a passive role rather than active participants in understanding the material (Shi & Zhao, 2024). However, the literature also cautions against treating AR as a default solution: applications built without regard for loading speed, tracking reliability, or content scalability tend to frustrate the same visitors they are meant to engage (Garcia, 2020). This supports a selective approach, AR reserved for specimens where three-dimensional visualization adds genuine educational value, rather than a requirement applied uniformly across an entire inventory.

Together, these three problems, fragile or manual inventory management, unstructured visitor communication, and under-selective exhibit engagement, define the rationale behind BioSphere's three core components: an offline-first web-based inventory system, a public-facing website with structured visitor requests and an FAQ Assistant, and QR-triggered AR limited to selected specimens.

Related Systems

Systems Supporting Offline-First, Structured Inventory Management

Several open-source and commercial collection management systems already address structured specimen inventory, though generally without an offline-first design. Symbiota (Gries et al., 2014) organizes

preserved plant, animal, fungal, and fossil records with taxonomic classification, specimen images, and data publishing to networks such as GBIF and iDigBio, but it is built for large-scale biodiversity data sharing rather than a single school-based museum. Arctos (Cicero et al., 2024) similarly integrates catalog records, media, and search across natural and cultural history collections, while Specify 7 (Specify Collections Consortium, 2018) adds query building, role-based permissions, and an optional public web portal. PastPerfect (PastPerfect Software, Inc., 1996), SkinSoft's S-Museum (SKINsoft, 2010), and Argus (Lucidea, 1988) extend collection management further into loans, exhibitions, donor tracking, and multi-user access, and Museum Space (Museum Space, 2019) adds modular conservation and curation workflows. Harapan and Widodo (2021) documented a smaller, more directly comparable case: a web-based inventory system built for the Jakarta Textile Museum specifically to replace an incomplete, Excel-based process, though it does not extend to specimen classification or AR. BioSphere will adopt only the elements appropriate to its MVP: caching selected specimen records, supporting offline search and viewing, and synchronizing newly created Uncataloged drafts. Offline editing of existing records, delta updates to existing records, and complex conflict-resolution mechanisms are excluded to reduce implementation risk and maintain a feasible capstone scope.

None of these systems, however, are designed to function without a live connection. Sirigiri et al. (2023) address this gap directly, showing that combining IndexedDB with service workers in an offline-first Progressive Web Application, using pre-caching, lazy loading, pagination, and delta synchronization, can cut initial load times by up to 40% and keep large datasets responsive even without network access, while noting that conflict resolution during resynchronization remains an open challenge. This offline-first pattern is what distinguishes BioSphere's architecture

from the collection management systems above: it adopts their approach to structured specimen records while adding the reliability guarantees that a school-based museum, without dedicated IT infrastructure, actually needs day to day.

For the BioSphere MVP, only the offline-first elements appropriate to the project's scope and development period will be adopted. These include caching selected specimen records for read-only viewing, supporting offline search and filtering, and synchronizing newly created Uncataloged text-only drafts after reconnection. Offline editing of existing records, delta updates to existing records, offline location or quantity changes, and complex conflict-resolution mechanisms are excluded to reduce implementation risk and maintain a feasible capstone scope.

Digitization workflow research reinforces how those records should be structured and captured. Abo Zeid (2022) proposed QR-code-based inventory as a low-cost way to speed up record access and reduce the accountability risks of poorly tracked storage. Ong et al. (2023) outlined a three-phase (preproduction, production, postproduction) digitization framework aimed at institutions with limited budgets and staff, and Seggi et al. (2024) demonstrated an image-to-data digitization workflow on a historically significant algal collection, with metadata organized around the Darwin Core standard. Inselect (Inselect, 2015), developed by the Natural History Museum, London, focuses specifically on the image-capture stage of that workflow, reaching throughput of up to 5,000 slides per day per person during a pilot project, though it is not itself a collection management system and must be paired with one.

Darwin Core, in particular, provides the standardized vocabulary, catalog number, scientific name, collection date, locality, habitat, and related terms, used to describe biodiversity occurrences and specimen records across institutions. BioSphere does not need to implement Darwin

Core as a full biodiversity-publishing system, but selected terms guide the design of its core specimen fields (Table 1), keeping the inventory scientifically organized and compatible with recognized documentation practice rather than functioning as a generic record database.

Table 1
Darwin Core Standard

BioSphere Specimen Field	Darwin Core-Aligned Concept
Accession or catalog number	catalogNumber
Scientific name	scientificName
Common name	vernacularName / local display field
Collector	recordedBy
Collection date	eventDate
Collection locality	locality
Habitat	habitat
Preservation or preparation methods	preparations
Notes/fun facts/ecological role	occurrenceRemarks and educational display fields

Systems Supporting Public-Facing Visitor Requests and Communication

A second cluster of research addresses how museums manage visitor-facing requests, which is directly relevant to BioSphere's public website. Jirapure et al. (2024) developed a web-based ticketing platform using QR codes for entry verification, reducing manual checking and supporting paperless transactions; Rathore et al. (2022) built a comparable e-ticketing system for Indian heritage sites, showing how QR codes can link an online transaction to a physical visit; and Anand et al. (2023) extended this further with Ticketless, which pairs QR-based entry with crowd prediction and identity verification. Parvathi et al. (2024) took a different approach, using a conversational Assistant interface for booking, visitor updates, and administrative analytics on booking and revenue.

These systems demonstrate the value of structured, trackable visitor interaction, but each includes features that exceed what the USC Biological Museum's expected usage, visit scheduling and curator meeting requests rather than high-volume ticket sales, actually requires. Full ticket purchasing and payment processing, crowd prediction, facial or biometric verification, and natural-language assistant booking all add complexity and, in some cases, privacy exposure that is not justified by the museum's current workflow. Ankor et al. (2024), studying reservation-system adoption more broadly, reached a similar conclusion: adoption depends as much on staff readiness, internet reliability, and cost as on the sophistication of the platform itself. Taken together, this literature supports BioSphere's narrower scope: structured web forms for visit and meeting requests, administrative review of request status, and a rules-based FAQ Assistant for repetitive visitor questions, deliberately excluding payment processing, crowd analytics, and biometric identification from the MVP.

Systems Supporting Selective, QR-Triggered AR

A third cluster addresses AR as an exhibit-engagement tool. Spadoni et al. (2022) developed AR clips for the Museo Astronomico di Brera in Milan to make abstract astronomical concepts easier to visualize, though without quantitative evidence isolating the AR intervention's impact. Shi and Zhao (2024) similarly found that AR annotations, scene recreation, and educational games help transform museum visitors from passive observers into active participants, while flagging tracking precision and interaction design as persistent limitations. Garcia (2020) built HARA, a mobile AR history application for Grade 6 students, and found it improved attitude and motivation toward the subject even without a measurable gain in test scores, while also surfacing slow loading times and limited content scalability as recurring technical constraints.

Consumer and education-focused AR applications illustrate simpler, more targeted implementations. Tuklas (Ateneo Laboratory for the Learning Sciences, 2019) uses camera-based sticker scanning to animate exhibits for young children at Museo Pambata; Animal Safari AR (LightUp, 2020) lets users place life-size 3D animal models in their environment alongside curriculum-relevant facts; and AssemblrWorld (Assemblr, 2018) provides a no-code platform for building AR-based lessons and exhibits. None of these are collection management systems, but they show that AR delivered through simple, targeted triggers, a sticker scan, a placed model, rather than a fully custom engine, is both achievable and effective for a small institution.

Read together, this literature supports treating AR as a focused enhancement rather than a default. BioSphere applies this by triggering AR only for selected specimen records through QR codes, so that three-dimensional visualization is used where it adds clear educational

value, rather than being built out across the entire inventory at the cost of loading speed, tracking reliability, and long-term maintainability.

Summary of Related Systems

Table 2 consolidates the systems reviewed above against thirteen features relevant to BioSphere's scope: specimen/collection cataloging, AR/3D visualization, taxonomic classification, media attachment, search and query tools, offline capability, role-based access control, report generation, web portal/publishing, digitization, location tracking, exhibition management, and FAQ assistant support. Each checkmark indicates that the corresponding system implements that feature, based on the literature and system documentation reviewed above.

Table 2
Summary of Related Systems

Systems ↓	Features →	1	2	3	4	5	6	7	8	9	10	11	12	13
Tuklas			✓											
Animal Safari AR			✓											
AssemblrWorld			✓							✓				
Symbiota		✓		✓	✓	✓			✓	✓				
Arctos		✓			✓	✓				✓				
Inselect				✓	✓						✓	✓		

Museum Space	✓			✓	✓			✓				✓	
Jakarta Textile Museum	✓						✓	✓					
Specify	✓		✓	✓	✓		✓	✓	✓		✓		
PastPerfect Museum Software	✓			✓				✓	✓			✓	
SkinSoft's S-Museum	✓	✓		✓			✓		✓		✓	✓	
Argus	✓			✓				✓	✓				
Proposed BioSphere	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓		✓

Key Features:

- | | |
|-------------------------------------|---------------------------|
| 1. Specimen/Collection Cataloging | 8. Report Generation |
| 2. AR/3D Visualization | 9. Web Portal/Publishing |
| 3. Taxonomic Classification | 10. Digitization |
| 4. Media Attachment | 11. Location Tracking |
| 5. Search & Query Tools | 12. Exhibition Management |
| 6. Offline Capability | 13. FAQ Chatbot |
| 7. Role-Based Access Control (RBAC) | |

The comparison shows a consistent pattern: systems built for collection management (Specify, SkinSoft's S-Museum, PastPerfect,

Symbiota, Arctos) cluster around cataloging, classification, search, and reporting, but none of them offer offline capability, AR visualization, or FAQ assistant support in combination. Systems built for AR engagement (Tuklas, Animal Safari AR, AssemblrWorld) do the opposite, offering visualization but no inventory, access control, or reporting functions at all. No single reviewed system combines structured cataloging, offline reliability, selective AR, and visitor-facing communication in one platform. BioSphere is the only system in the comparison that supports twelve features, and it is particularly distinguished by its offline capability, a feature entirely absent from every other collection management or ticketing system reviewed, despite being one of the more consistently cited needs in the literature.

CHAPTER 3

TECHNICAL BACKGROUND

This chapter presents the technical foundation of the BioSphere system, covering its architecture, backend technology stack, database integration, security measures, file and media handling, deployment infrastructure, development tools, quality assurance, and the technical considerations that informed the system's design.

Integrated Development Environment (IDE)

An Integrated Development Environment (IDE) is a software application that provides developers with a unified environment for writing, editing, debugging, and testing source code. It integrates essential development tools such as a code editor, compiler or interpreter support, debugging utilities, and project management features into a single interface, improving development efficiency and software quality (Microsoft, 2024). An IDE will be used throughout the development of BioSphere to facilitate efficient coding, debugging, testing, and maintenance of the system.

Planning and Project Management Tools

Planning and project management tools are applications that assist development teams in organizing tasks, assigning responsibilities, monitoring project progress, and coordinating activities throughout the software development lifecycle (Atlassian, 2024). These tools improve collaboration, resource allocation, and project scheduling, helping ensure that development milestones are completed within the planned timeframe. Such tools will be used to organize and monitor the development activities of BioSphere from planning to deployment.

Design and Prototyping Tools

Design and prototyping tools enable developers and designers to create wireframes, interface layouts, and interactive prototypes before software implementation. These tools support visualization, usability evaluation, and collaborative refinement of user interface designs, allowing potential issues to be identified early in the development process (Figma, 2024). These tools will be used to design and validate the user interface and user experience of BioSphere before implementation.

Frontend Development

Frontend development refers to the process of creating the graphical user interface of a web application that users interact with through a web browser. It focuses on developing responsive, interactive, and user-friendly interfaces while ensuring compatibility across different devices and screen sizes (MDN Web Docs, 2024). Front-end development technologies will be used to build the user interfaces for both the museum management system and the public-facing website of BioSphere.

Backend Development

Backend development involves creating the server-side components of a software application responsible for processing business logic, handling user requests, managing authentication, and communicating with databases. It serves as the core processing layer that enables system functionality and secure data management (MDN Web Docs, 2024). Back-end technologies will be used to implement the core functionalities of BioSphere, including inventory management, visitor requests, authentication, and data processing.

Programming Languages

A programming language is a formal language used to write instructions that enable computers to perform specific tasks. Modern programming languages provide features that improve software readability, maintainability, and scalability while supporting the development of reliable applications (Microsoft, 2024). Programming languages will be used to implement the front-end, back-end, and supporting functionalities of BioSphere.

Database Management

A Database Management System (DBMS) is software that enables the storage, organization, retrieval, and management of structured data while maintaining data integrity, consistency, and security (PostgreSQL Global Development Group, 2024). DBMSs provide mechanisms for efficient querying, updating, and maintaining large volumes of information required by information systems. A database management system will be used to store and manage the specimen inventory, user accounts, visitor requests, inquiries, reports, and other system data of BioSphere.

Backup and Recovery

Backup and recovery technologies protect system information by creating restorable copies of databases and other essential system resources. PostgreSQL provides database export and archival utilities such as `pg_dump`, while `pg_restore` reconstructs a database from supported archive formats (PostgreSQL Global Development Group, 2026a, 2026b). In BioSphere, controlled backup and recovery procedures will be used to protect the relational database and other essential system data. Backup files will be access-controlled, stored in an approved location, and periodically tested to determine whether the system can be restored following accidental deletion, data corruption, deployment failure, or another data-loss event.

Spreadsheet and Comma-Separated Values Data Import and Export

Spreadsheet and Comma-Separated Values (CSV) processing technologies allow structured records to be transferred between information systems and commonly used tabular file formats. CSV is a widely used format for exchanging tabular data and is formally associated with the text/csv media type (Shafranovich, 2005). In BioSphere, spreadsheet and CSV processing will support the controlled migration of existing museum specimen records into the system. Imported rows will undergo field mapping, required-field checking, data-type validation, format validation, and possible duplicate-record checking before they are accepted. The system will also allow authorized curators to export approved inventory and report data for institutional documentation and analysis.

Object-Relational Mapping

Object-Relational Mapping (ORM) is a software development technique that enables applications to interact with relational databases through programming language objects rather than manually writing database queries. This approach simplifies database operations, improves code maintainability, and reduces development complexity (Prisma, 2024). An ORM will be used to facilitate efficient communication between the application's data models and the database in BioSphere.

Progressive Web Application and Service Worker Technology

A Progressive Web Application (PWA) is a web application that uses modern browser capabilities to provide a more reliable and application-like experience across supported devices. A service worker operates between the web application, browser, and network, allowing it to intercept network requests, cache selected resources, and support application functionality when a network connection is temporarily unavailable (Mozilla, 2026a). In BioSphere, PWA and

service worker technologies will support the offline-first curator application by caching essential application resources and enabling selected inventory functions to remain accessible during temporary internet interruptions. Public website submissions, external notifications, and other connectivity-dependent functions will continue to require an internet connection.

Offline-First Web Applications

Offline-first web applications are designed to continue functioning even when internet connectivity is temporarily unavailable by storing selected application resources and data locally before synchronizing changes once connectivity is restored (MDN Web Docs, 2024). This approach improves system reliability and ensures uninterrupted access to essential functionalities despite network interruptions. This approach will be applied to BioSphere to allow authorized inventory operations to continue during temporary internet outages and synchronize changes once connectivity becomes available.

Client-Side Data Storage

Client-side data storage allows a web application to retain structured information within the user's browser. IndexedDB is a browser-based application programming interface designed to store significant amounts of structured data, including files and binary objects, while supporting indexes for efficient data retrieval (Mozilla, 2025a). It can persist information in the browser and support applications that operate regardless of temporary network availability (Mozilla, 2025a). In BioSphere, IndexedDB or an equivalent client-side storage technology will retain selected cached specimen records and new Uncataloged text-only drafts during temporary internet interruptions.

Authentication and Role-Based Access Control

Authentication is the process of verifying the identity of a user before granting access to a system, while role-based access control (RBAC) restricts

system functions based on predefined user roles and permissions. Together, these mechanisms help protect sensitive information and ensure that users can only access resources appropriate to their responsibilities (NIST, 2020). These security mechanisms will be implemented in BioSphere to control access to inventory records, administrative functions, and public resources according to user roles.

Augmented Reality

Augmented Reality (AR) is a technology that overlays digital information, such as images, animations, or three-dimensional models, onto the real-world environment, allowing users to interact with virtual content within their physical surroundings (Azuma, 1997). AR has become increasingly popular in museums as an educational tool that enhances visitor engagement and learning experiences. In BioSphere, AR will be used as an optional exhibit enhancement for selected specimens to provide interactive three-dimensional educational content.

Controlled Knowledge-Based Frequently-Asked-Questions Assistant

A controlled knowledge-based Frequently-Asked-Questions (FAQ) assistant is a conversational interface that retrieves responses from a predefined and authorized collection of questions and answers. Unlike an unrestricted generative artificial intelligence system, it is limited to information contained in its approved knowledge base. In BioSphere, the public website FAQ assistant will provide answers only from museum information reviewed and approved by the curator. When no approved answer is available, the assistant will not generate or invent a response and will instead direct the visitor to the General Inquiry form. The assistant will not approve visit requests, modify museum records, or access unpublished internal inventory information (BioSphere Software Requirements Specification, 2026).

Quick Response Code Technology

Quick Response (QR) codes are two-dimensional barcodes capable of storing information that can be rapidly accessed using devices equipped with cameras. They provide a convenient method for linking physical objects to digital resources, enabling users to retrieve information quickly without manually entering web addresses (ISO/IEC 18004, 2015). QR code technology will be used in BioSphere to provide visitors with quick access to approved digital exhibit information for selected museum specimens.

Representational State Transfer Application Programming Interface Design and Testing

A Representational State Transfer (REST) Application Programming Interface (API) is an architectural style for designing web services that enable communication between software applications using standard HTTP methods and stateless interactions (Fielding, 2000). REST APIs promote interoperability, scalability, and maintainability by allowing clients and servers to exchange data efficiently through standardized requests and responses. A RESTful API architecture will be used in BioSphere to facilitate secure and efficient communication between the user interface, server-side components, and the system database.

Version Control, Continuous Integration and Continuous Deployment

Version control is a software development practice that records and manages changes to source code over time, enabling developers to collaborate, track revisions, and restore previous versions when necessary. Continuous Integration (CI) and Continuous Deployment (CD) are development practices that automate the building, testing, and deployment of software, helping maintain code quality and streamline software delivery (GitHub, 2024). These practices will be applied during the development of BioSphere to manage source code

changes, support collaborative development, and facilitate efficient integration and deployment throughout the project lifecycle.

Containerization and Deployment

Containerization is a software deployment approach that packages an application together with its required libraries, dependencies, and configuration files into isolated containers that can run consistently across different computing environments (Docker, 2024). This approach improves portability, scalability, and deployment consistency by minimizing differences between development, testing, and production environments. Containerization and deployment technologies will be utilized to simplify the deployment, maintenance, and future scalability of BioSphere across supported environments.

Cloud Object/File Storage Services

A cloud object/file storage service is a managed cloud-based solution that stores and retrieves unstructured data such as images, documents, videos, and other digital files through internet-based services (Supabase, 2024). Unlike traditional databases, object storage is optimized for handling large binary files while providing scalability, durability, and secure access. In BioSphere, these tools will be utilized to securely store specimen photographs, supporting images, and other multimedia resources associated with biological records, allowing users to efficiently upload, retrieve, and manage digital assets without overloading the primary relational database.

AI-Powered 3D Model Generation

An AI-powered 3D model generation platform uses artificial intelligence (AI) to automatically create three-dimensional models from text prompts, images, or other forms of input, significantly reducing the time and expertise required for traditional 3D modeling (Tripo, 2024). These platforms employ machine learning techniques to generate realistic geometric structures and textures suitable for

visualization and interactive applications. In BioSphere, Tripo 3D is used to generate initial 3D representations of biological specimens, enabling the museum to provide interactive digital visualizations that enhance educational and research experiences.

AI-Powered 3D Asset Generation (Text/Image-to-3D)

An AI-powered 3D asset generation platform converts text descriptions or two-dimensional images into editable three-dimensional assets using generative artificial intelligence (Meshy, 2024). These tools accelerate the creation of detailed 3D models by automating mesh generation, texturing, and refinement processes, making them valuable for digital content creation and visualization. This will be utilized to transform specimen images into enhanced 3D assets that can be further refined for use in the system's interactive specimen viewer, reducing the manual effort required to develop high-quality digital models.

Quality Assurance and Test Management

A Quality Assurance (QA) and Test Management platform is a software solution designed to organize, execute, and monitor software testing throughout the development lifecycle (Qase, 2024). It enables development teams to create and manage test cases, track test execution results, document defects, and generate reports to ensure that software meets its functional and quality requirements before deployment. For the development of BioSphere, this will be used to develop and manage test cases, record test outcomes, monitor defects, and verify that the system's features—including specimen management, user authentication, and 3D visualization—function correctly and satisfy the project's specified requirements.

CHAPTER 4

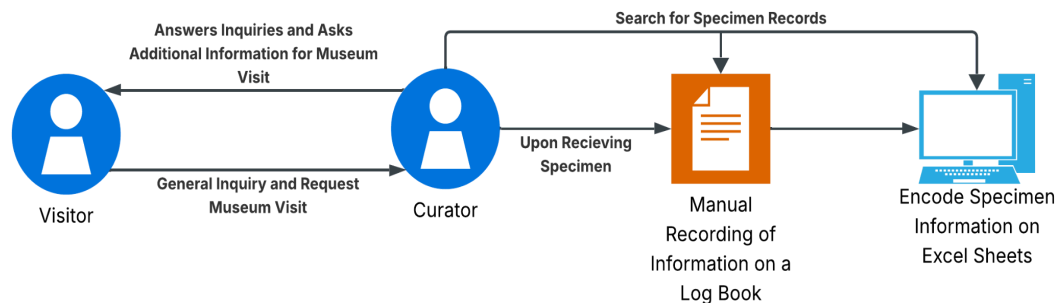
DESIGN AND METHODOLOGY

This chapter describes the methodology adopted for the development of BioSphere. It presents the project's conceptual framework, system analysis and design using use case diagrams (UCDs) and entity-relationship diagrams (ERDs), the software development model and development approach, the software development tools used and project management and procedures followed throughout the design, implementation, and evaluation of the proposed system.

4.1 Conceptual Framework

Figure 1

Manual Process

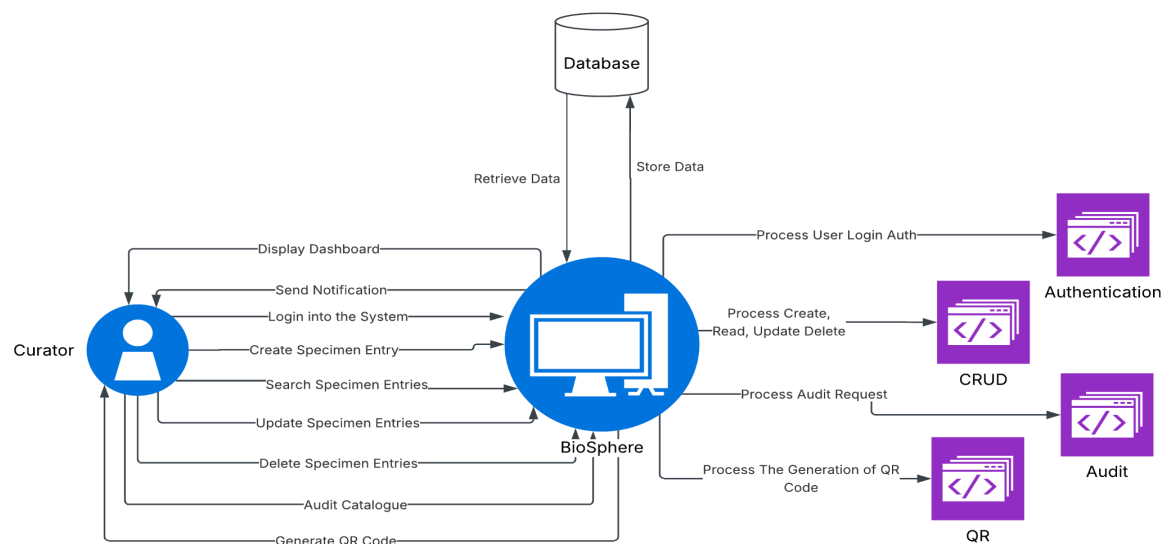


As illustrated above, Figure 1 shows the current manual process of operations of USC-BM. Upon receiving a new specimen, the curator records its essential information, such as the accession number, scientific name, common name, collector's name, date collected, and collection location, in a physical logbook before manually encoding the verified information into a Microsoft Excel spreadsheet for inventory management. When specimen information is incomplete or requires verification, the curator consults external taxonomy references before updating the records. In addition to inventory responsibilities, the curator also manages visitor inquiries and museum visit requests through manual communication channels, such as email and text messages. Information

required for campus entry is collected manually, submitted through the university's existing visitor management procedures, and followed up until approval is received.

While these manual practices allow the museum to maintain its daily operations, they create several operational challenges. The reliance on paper-based documentation and spreadsheet encoding increases the risk of data loss, duplication, encoding errors, and the absence of a reliable backup and recovery mechanism. Likewise, handling visitor inquiries and requests through multiple communication channels requires continuous manual coordination, resulting in additional administrative workload, delayed responses, and a greater possibility of miscommunication. These limitations highlight the need for a centralized digital system capable of streamlining specimen inventory management, improving visitor request coordination, and providing secure data management with backup and recovery capabilities.

Figure 2
Curator Conceptual Framework



As shown in figure 2, it illustrates the conceptual framework of the curator's interaction with BioSphere. As the primary internal user of the system, the curator accesses BioSphere by authenticating through the login module before gaining access to the system dashboard. Once authenticated, the curator can perform specimen inventory management functions, including creating, searching, updating, cataloging, and archiving specimen records. These operations are processed through the system's Create, Read, Update, and Delete (CRUD) functions, while all authorized actions are securely stored and retrieved from the centralized database.

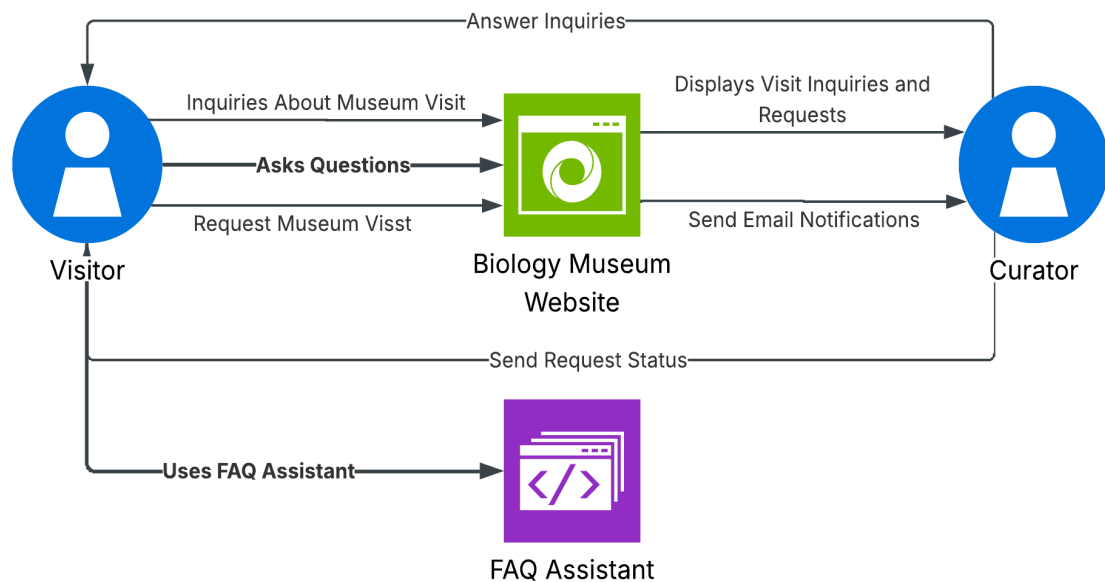
In addition to inventory management, the curator may generate QR codes for specimen records, should they be used as exhibits and are to be linked to their corresponding public exhibit pages. The system also maintains audit logs to record significant user activities, supporting accountability and traceability of inventory-related operations. Throughout the curator's interaction with the system, BioSphere processes authentication requests, inventory transactions, QR code generation, and audit logging while managing the storage and retrieval of data from the centralized database. By integrating these core functions into a single platform, BioSphere streamlines the curator's daily responsibilities, reduces reliance on manual recordkeeping, and improves the efficiency, security, and maintainability of the museum's specimen inventory management process.

As illustrated by figure 3 below, it shows the conceptual framework of the visitor's interaction with the Biology Museum Website. Visitors may submit general inquiries or museum visit requests through the website, which serves as the primary communication platform between visitors and the curator. Submitted General Inquiries and Visit Requests are displayed to the curator for review. BioSphere sends the authorized curator an email notification when a new submission is received. The curator may compose an inquiry reply or Visit Request update through BioSphere, after which the system sends the message to the visitor's email address. The email function is outbound only. Replies sent by visitors to the museum's official email address are handled outside BioSphere

and are not automatically imported or synchronized. The curator may manually update the corresponding record when necessary.

Figure 3

Visitor Conceptual Framework



As shown below in figure 4, it illustrates the restricted technical interaction of the Developer with the BioSphere system. The process begins when the developer accesses BioSphere through the approved authentication mechanism. Once authenticated, the developer may perform only limited and formally authorized functions, including onboarding the initial curator through a secure email invitation and password-setup process, disabling or removing curator access when authorized by the museum or university, and deploying or updating approved augmented reality assets. BioSphere processes these requests through the corresponding curator account provisioning and AR asset deployment functions while retrieving and storing the required information in the database. These controls ensure that developer access remains limited to technical administration and does not extend to routine museum operations such as specimen management, inquiry processing, Visit Request handling, reporting,

FAQ-knowledge management, or QR-exhibit content management. Approved maintenance changes to the static public website are performed through the frontend source code and deployment process rather than through the restricted Developer interface.

Figure 4
Developer Conceptual Framework

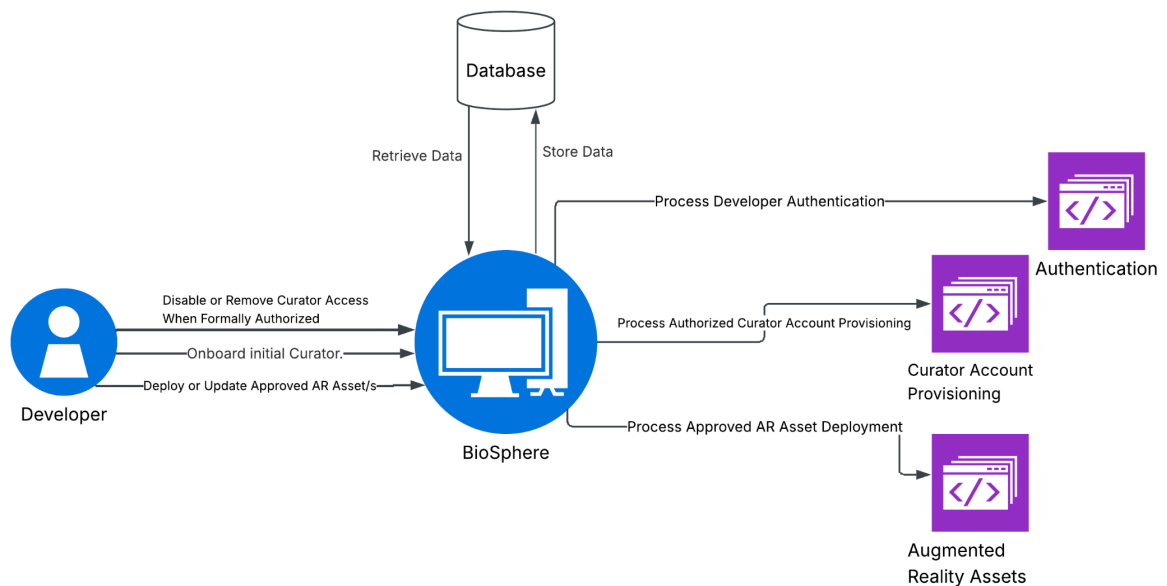
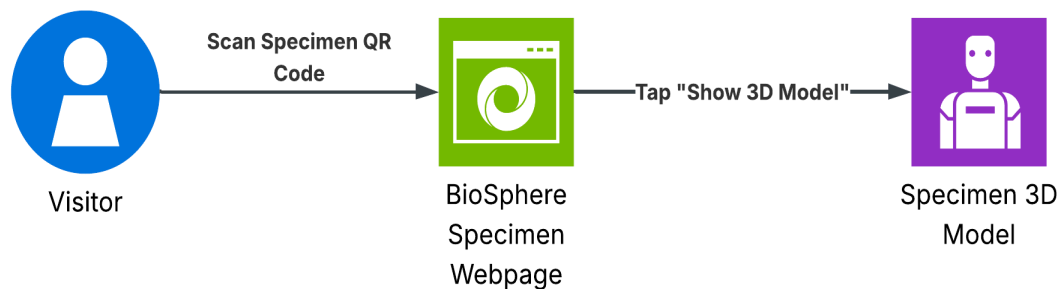


Figure 5
BioSphere Proposed Augmented Reality Process



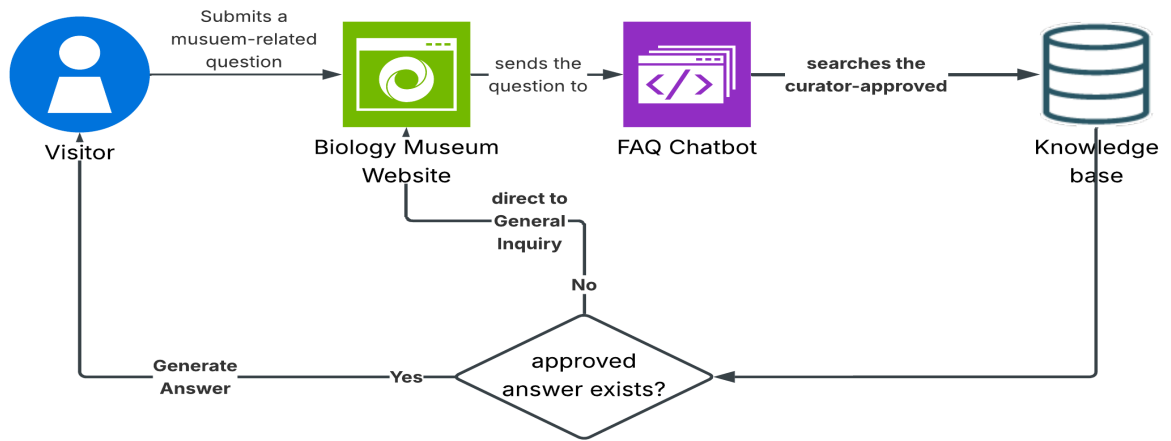
As illustrated in Figure 5, the proposed augmented reality (AR) process of BioSphere begins when a visitor scans the QR code attached to a

curator-selected specimen, which opens the corresponding exhibit page containing approved educational information. Specimens for AR presentation are selected by the curator based on the intended research or educational objective and may consider conservation status, rarity, endemism, exotic or invasive status, and economic or ecological value. These serve as selection guidelines rather than a fixed scoring system. For specimens with approved three-dimensional assets, compatible devices may launch the AR/3D experience through the exhibit page. If QR scanning or AR is unsupported, visitors may use the human-readable direct URL on the exhibit label and continue accessing the standard exhibit page and its non-AR content.

As shown below in figure 6, it illustrates the process of the Controlled Knowledge-Based FAQ Assistant integrated into the BioSphere public website. A visitor submits a museum-related question through the website, which forwards the question to the FAQ Assistant. The assistant then searches the curator-approved knowledge base for a relevant response. If an approved answer is available, the assistant displays the corresponding answer to the visitor. If no relevant approved answer is found, the assistant informs the visitor and directs them to the General Inquiry form for further assistance. This process provides visitors with faster access to common museum information while ensuring that responses are limited to curator-approved content. The FAQ Assistant does not submit, approve, decline, or manage museum visit requests.

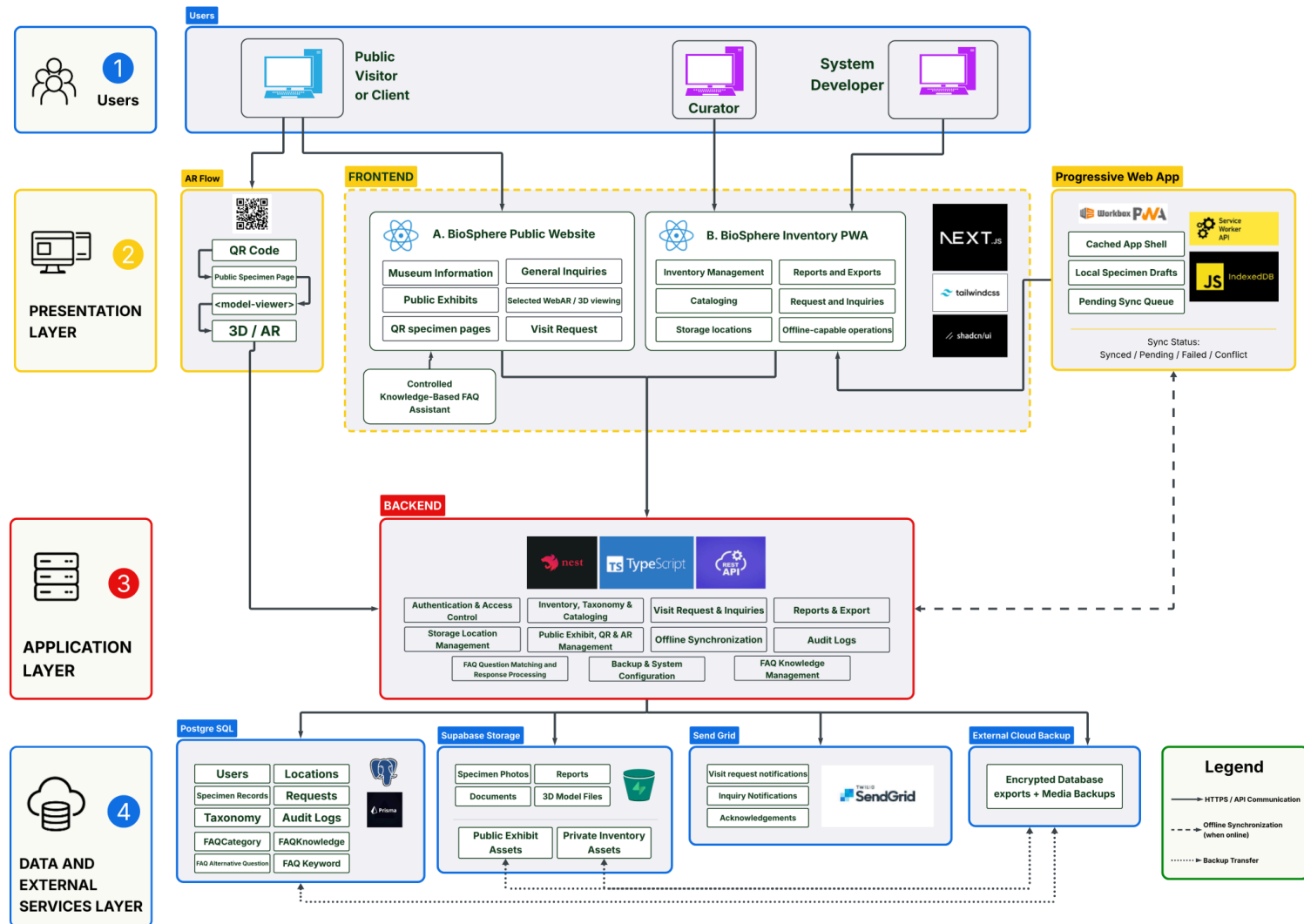
Figure 6

BioSphere Proposed Controlled-Knowledge FAQ Assistant Process



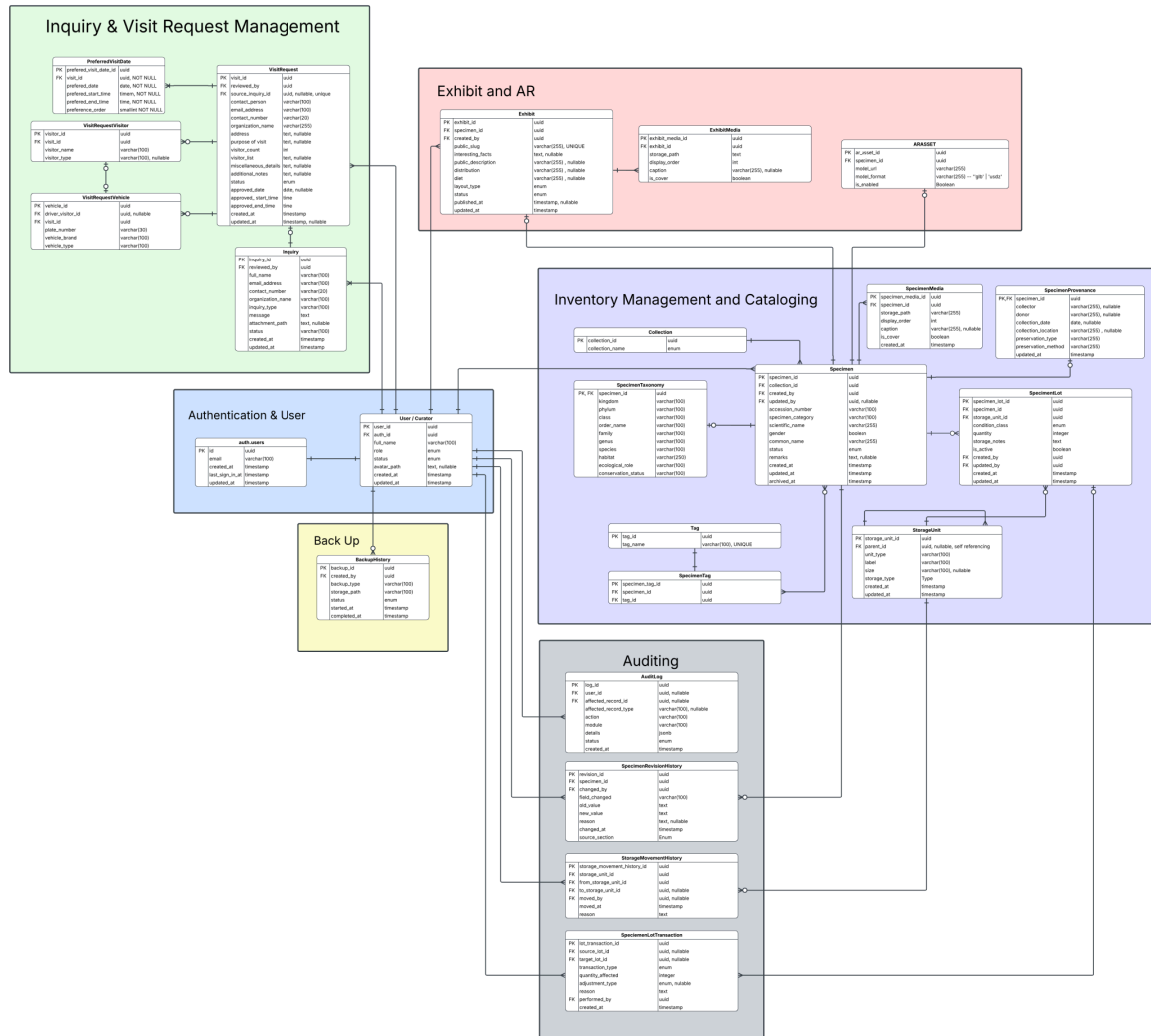
As shown in Figure 7, the BioSphere architectural framework integrates the presentation, application, data, and external services layers of the proposed system. It supports three primary users: public visitors, the museum curator, and the developer. Public visitors access approved museum information, selected exhibits, inquiries, visit requests, the Controlled Knowledge-Based FAQ Assistant, and QR-linked exhibit pages with optional WebAR content. The curator uses the Inventory Progressive Web Application to manage specimens, cataloging, storage locations, reports, requests, inquiries, QR exhibits, and FAQ knowledge, while the developer is limited to authorized technical functions such as curator onboarding, account removal, and AR asset deployment. The backend manages authentication, inventory, taxonomy, storage, visitor services, reports, QR exhibits, offline synchronization, audit logging, backups, and FAQ processing. System data is stored in PostgreSQL, while Supabase Storage manages images, documents, exhibit assets, and 3D models. SendGrid is used only for authorized outbound email notifications, replies, visit-request updates, and account-security messages. PWA features also provide cached resources, read-only specimen caching through IndexedDB, and synchronization of newly created offline Uncataloged drafts.

Figure 7
BioSphere Architectural Framework



4.2 Analysis and Design

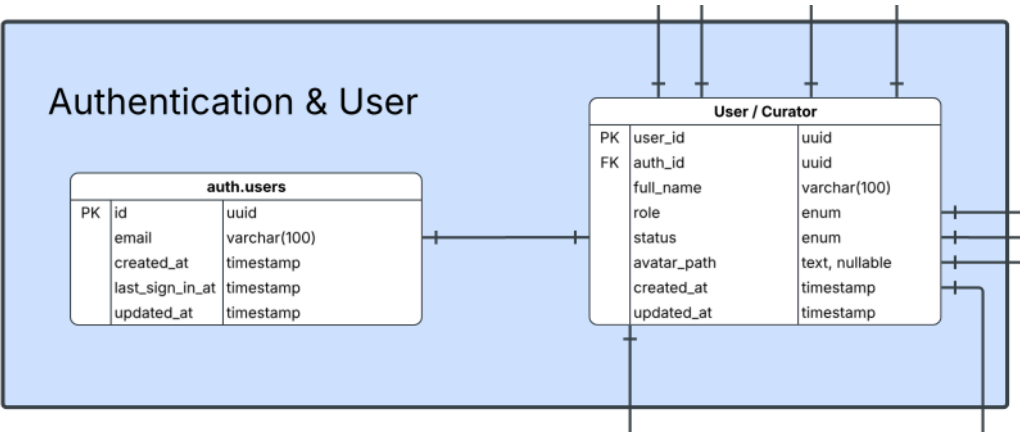
Figure 8
Entity Relationship Diagram Full View



As shown in Figure 8, the BioSphere database is organized into six major clusters: Authentication and User, Inquiry and Visit Request Management, Exhibit and AR, Inventory Management and Cataloging, Auditing, and Backup. The Specimen table serves as the main inventory record, while SpecimenLot stores each subgroup's quantity, condition, and location. The User/Curator table connects across the system to support accountability for major actions. This

clustered design separates each functional area while preserving the necessary database relationships.

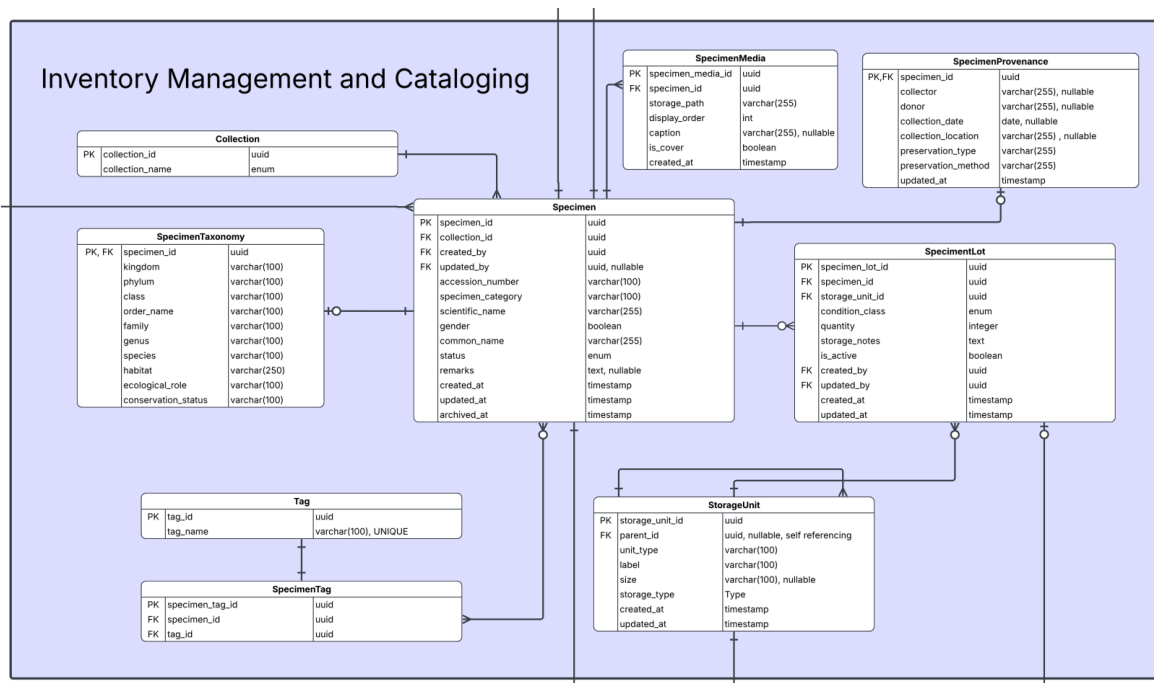
Figure 9
Authentication and Users Tables



As shown in Figure 9, the Authentication and Users segment manages staff identity and system access. The `auth.users` table, managed by Supabase, stores login credentials and has a one-to-one relationship with the `User/Curator` table, which stores the staff member's profile, role, and account status. Other tables reference the `User/Curator` table to identify the staff member responsible for system actions. The `BackupHistory` table also references the user who initiated each backup operation.

Figure 10

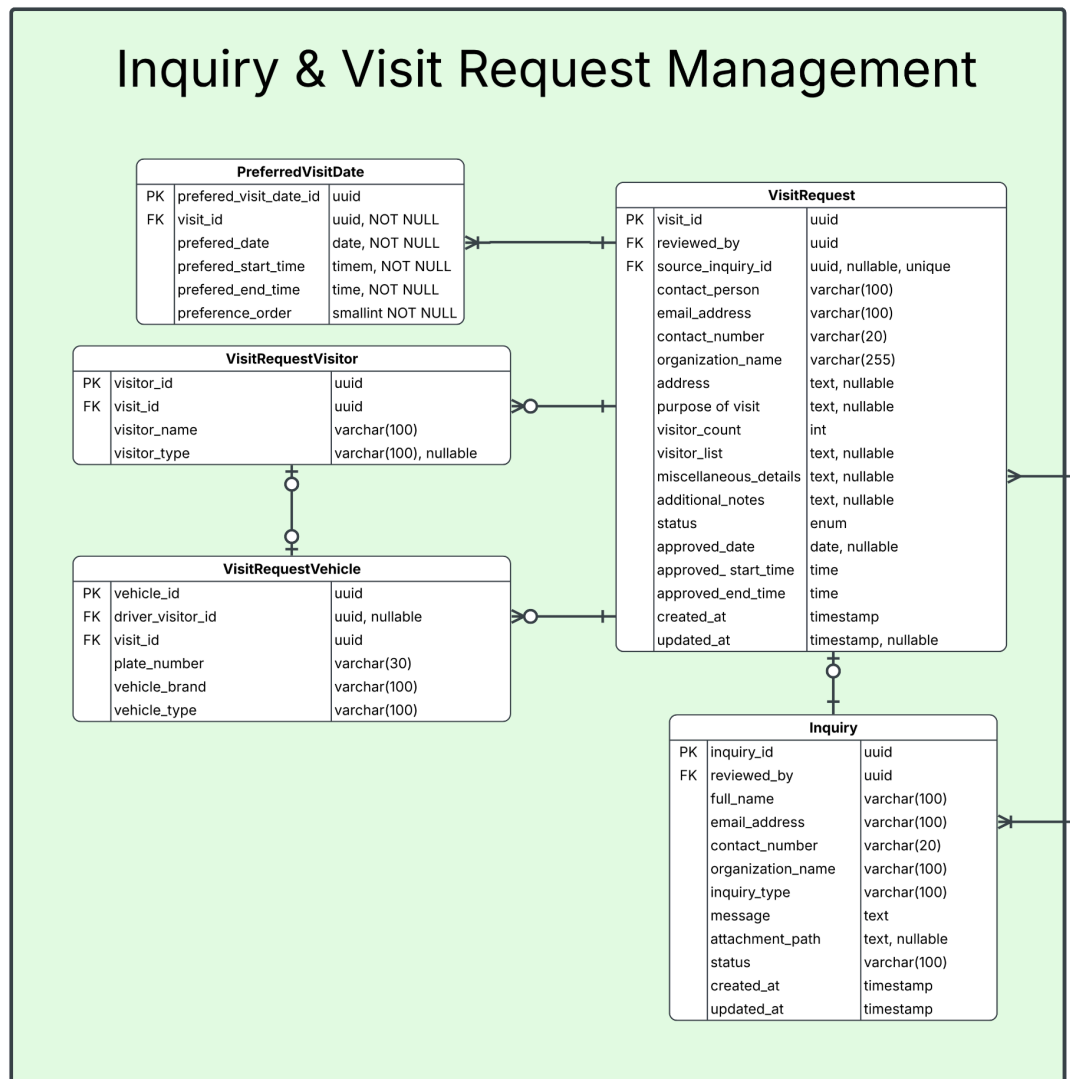
Inventory Management and Cataloging Tables



As shown in Figure 10, the Inventory Management and Cataloging segment is centered on the Specimen table, which stores information shared by the complete specimen record, such as the accession number, category, scientific name, common name, gender, and cataloging status. Supporting tables include Collection, SpecimenTaxonomy, SpecimenProvenance, and SpecimenMedia, which store the specimen's collection grouping, classification, provenance, and media information. The SpecimenLot table stores the quantity, condition class, and storage location of each physical subgroup associated with a specimen. This allows one specimen record to have multiple lots when its items are stored in different locations or have different conditions. The specimen's total quantity is obtained from the combined quantities of its lots. The StorageUnit table represents the museum's hierarchical storage locations, while the Tag and SpecimenTag tables allow specimens to be assigned flexible, curator-defined keywords for easier searching and organization.

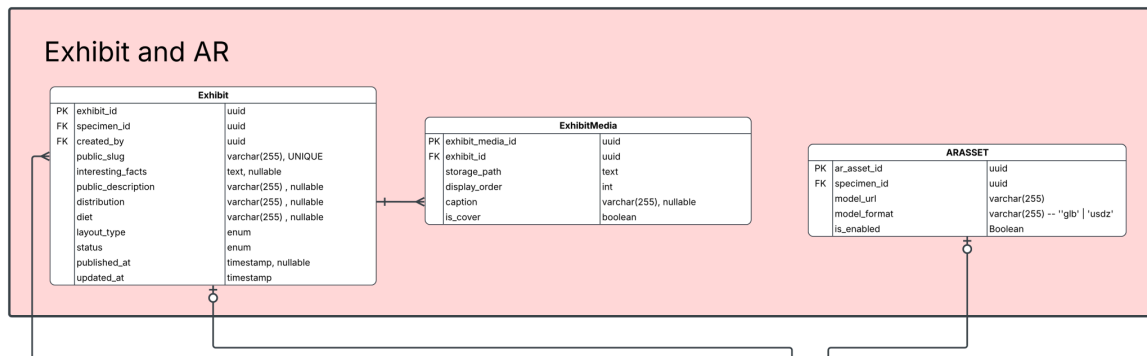
Figure 11

Inquiry and Visitor Request Management Tables



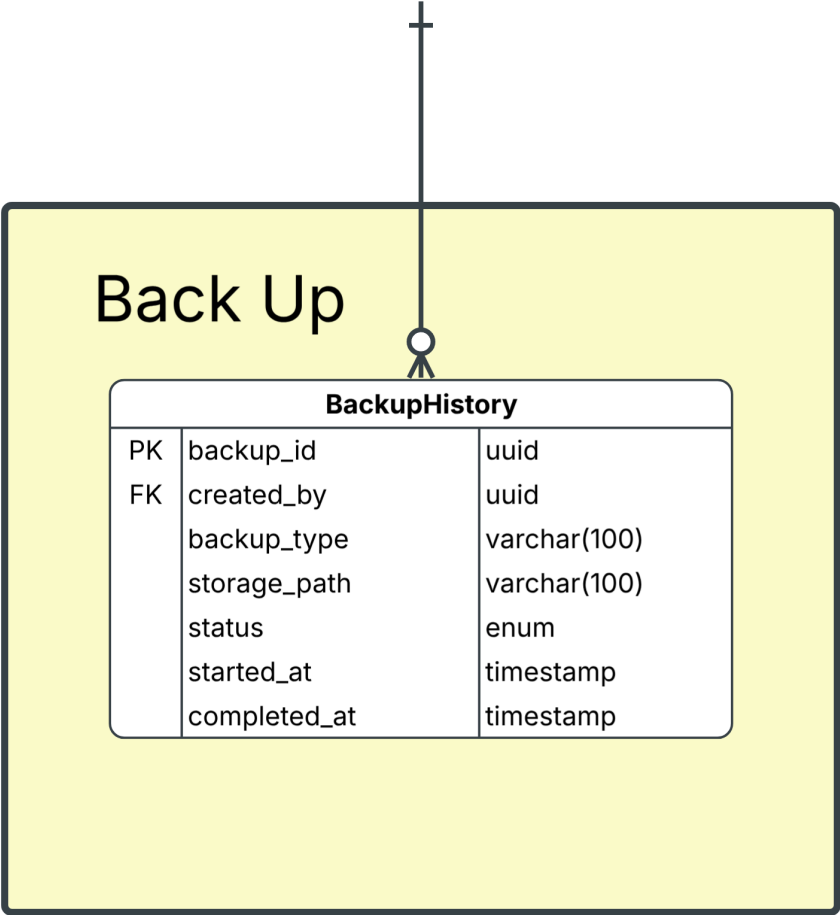
As shown in Figure 11, it illustrates the structure of the Inquiry and Visit Request Management segment, which handles public inquiries and museum visit requests. The Inquiry table captures general contact submissions, which may optionally lead to a formal VisitRequest. The VisitRequest table is supported by PreferredVisitDate, VisitRequestVisitor, and VisitRequestVehicle, allowing requesters to propose multiple dates and track visitors and vehicles individually. This normalized structure replaced earlier flat text fields, enabling more accurate tracking of group visits.

Figure 12
Exhibit and AR Tables



As shown in Figure 12, the Exhibit and AR segment manages public-facing exhibit content and augmented reality assets associated with approved specimens. The Exhibit table is created only when a specimen is approved for public display and stores its QR code value and visitor-facing description. ExhibitMedia stores the curated images presented on the exhibit page, while ARAsset stores the approved three-dimensional model file and format used for AR viewing. Both Exhibit and ARAsset maintain a one-to-zero-or-one relationship with Specimen because not every specimen is required to have a public exhibit or AR experience. The system does not automatically rank or determine specimens for AR presentation. AR selection remains under curator discretion and may be guided by the specimen's relevance to a research or educational objective, conservation status, rarity, endemism, exotic or invasive status, and economic or ecological value. Once a specimen and its AR presentation have been approved, the corresponding AR asset may be associated with the selected exhibit.

Figure 13
Back Up Table

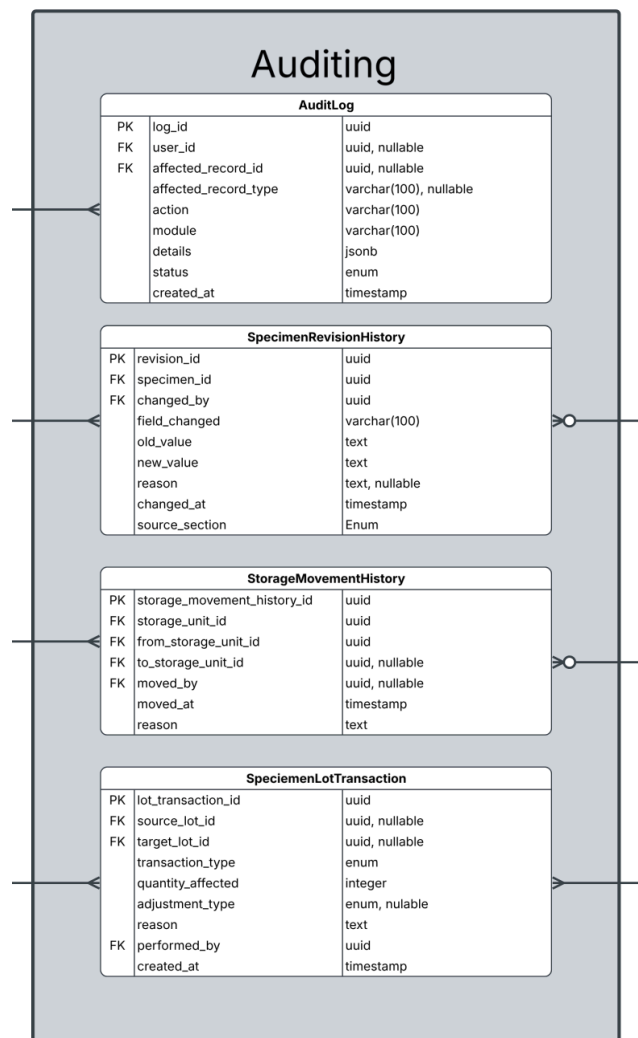


As shown above, figure 13 illustrates the structure of the Back Up segment, which supports the Authentication and User cluster by tracking system backup activity. The BackupHistory table records each backup event, including its type, storage path, and status, along with its start and completion timestamps. Each backup entry is linked to the staff member who initiated it through a reference to the User/Curator table. This allows administrators to monitor backup history and verify that data protection processes are being carried out consistently.

As shown in Figure 14, the Auditing segment tracks system activities and inventory changes for accountability. The AuditLog table records general staff

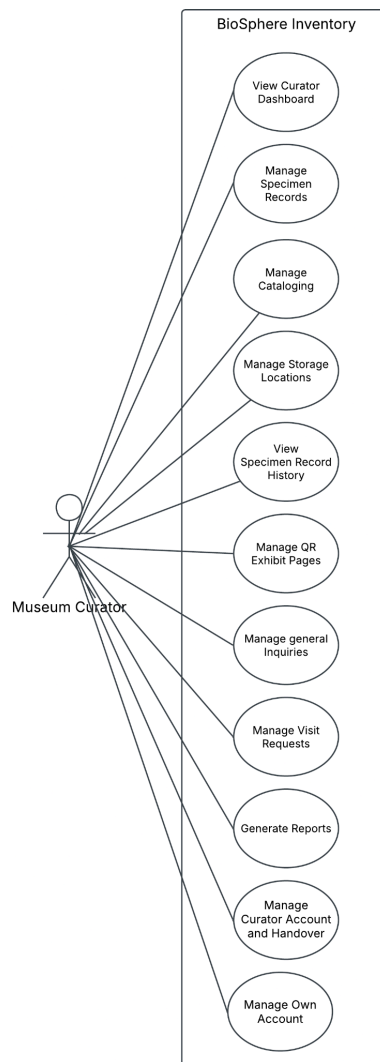
actions, while SpecimenRevisionHistory tracks specimen-field changes. StorageMovementHistory records storage-unit relocations, and SpecimenLotTransaction records lot movements, condition changes, splits, merges, and quantity adjustments. Together, these tables keep major actions traceable by recording the responsible user, timestamp, and reason where applicable.

Figure 14
Auditing Tables



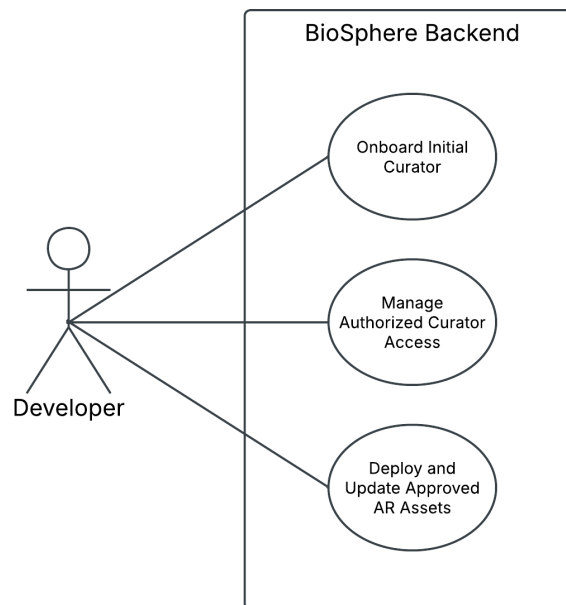
As shown below, figure 15 presents the functions available to the Museum Curator within the authenticated BioSphere Inventory application. The curator can access the dashboard, manage cataloged, uncataloged, and archived specimen records, organize specimen storage locations, review specimen history, manage QR exhibit pages, process general inquiries and visit requests, generate reports, and manage curator account onboarding and succession. These functions represent the museum's primary inventory, exhibit-content, and visitor-coordination activities.

Figure 15
Curator Use Case Diagram



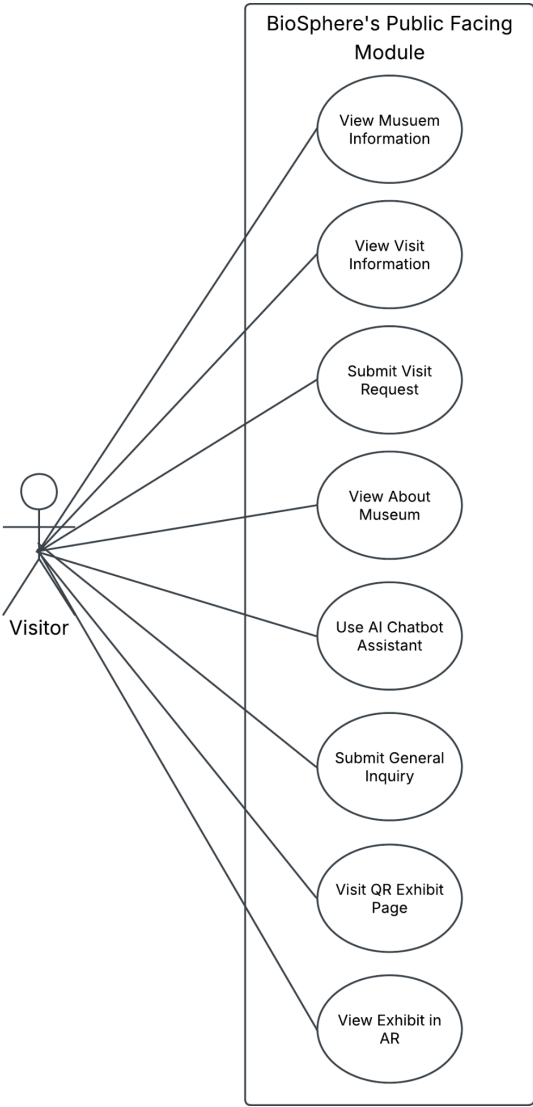
As illustrated in Figure 16, the Developer is limited to authorized technical functions within the BioSphere backend. These functions include onboarding the initial curator, managing curator access when formally authorized, and deploying or updating approved AR assets for selected exhibits. The Developer cannot access routine museum operations, including specimen records, storage locations, inquiries, visit requests, reports, FAQ knowledge, or QR exhibit content.

Figure 16
Developer Use Case Diagram



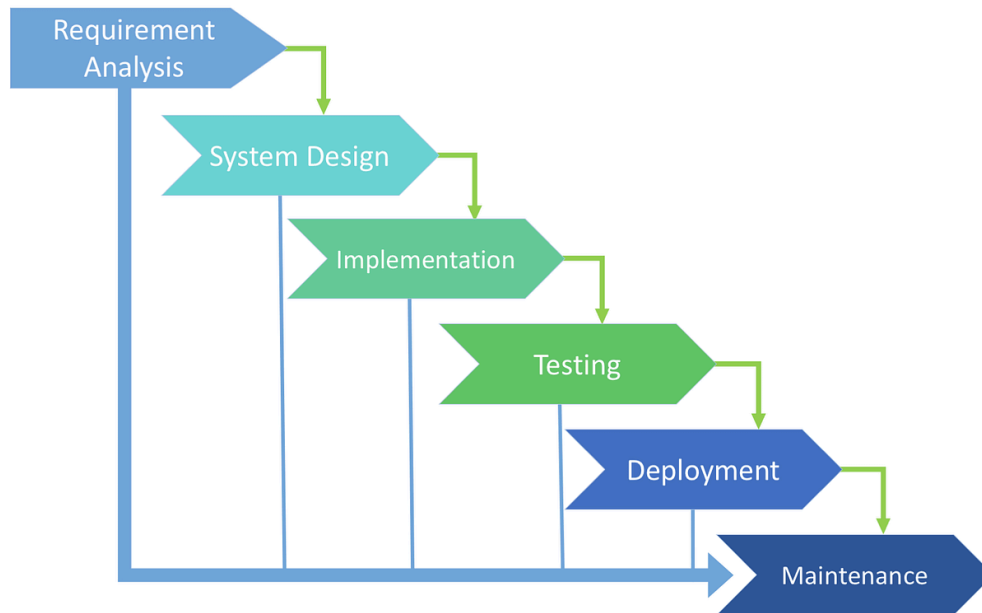
As depicted below, Figure 17 presents the services accessible to public visitors through BioSphere's public-facing application. Visitors can view museum and visit information, learn about the museum, submit visit requests and general inquiries, access QR-based exhibit pages, and view selected exhibits through web-based augmented reality. The Controlled Knowledge-Based FAQ Assistant allows visitors to submit supported museum-related questions and receive responses based only on active curator-approved knowledge. When no approved answer is available, the assistant directs the visitor to the General Inquiry form.

Figure 17
Visitor Use Case Diagram



4.3 Development Model

Figure 18
Waterfall Development Model



As shown in Figure 18, the development of the proposed BioSphere system will follow the Waterfall Development Model, is a sequential software development methodology in which each phase is completed before proceeding to the next (Atlassian, n.d.). The model provides a structured process that allows the development team to establish clear objectives, complete the required documentation, and thoroughly verify each phase before moving forward. Since the major functional and non-functional requirements have been identified through client consultations and documented before implementation, the Waterfall Model provides a structured framework for developing BioSphere. Limited validation and approved revisions may still occur before each phase is finalized.

The Waterfall Development Model is selected for this capstone project as the system has well-defined requirements and a clearly established scope. Each development phase produces deliverables that are reviewed and approved before proceeding to the next stage, minimizing the need for significant revisions during implementation. This structured approach ensures that the system is developed consistently according to the agreed specifications. Any additional features or enhancements identified after project completion may be considered in future versions rather than being incorporated into the current implementation.

System Design

After the requirements are analyzed and approved, the team proceeds to system design by creating database schemas, architecture diagrams, UI prototypes, module specifications, and workflows. These outputs serve as the blueprint for implementing and integrating the system's modules. They also help identify potential design issues before development begins.

Implementation

The implementation phase involves translating the approved system design into executable software. During this phase, the development team develops the different modules of BioSphere, including Specimen Inventory Management, Museum Visit Request Management, General Inquiry Management, Controlled Knowledge-Based FAQ Assistant, QR Exhibit Management, Reports and Data Export, Audit Logging, Backup and Limited Offline Draft Synchronization, Curator Account Management, and selected WebAR functionalities. Each module is implemented according to the approved design specifications and is subjected to preliminary testing to ensure correct functionality.

Testing

Once implementation is completed, the system undergoes comprehensive testing to verify that all modules function according to the documented

requirements. This phase includes verifying individual components, testing the integration of related modules, identifying software defects, and ensuring that the system performs as expected. Corrections are made for any identified issues before the system proceeds to deployment.

Deployment

After successful testing, the completed BioSphere system is deployed for use by the intended users. The finalized system is installed and configured within the museum environment, allowing museum curators, developers, and public visitors to access the functionalities according to their respective roles. User acceptance is also conducted to confirm that the implemented system satisfies the approved project requirements.

Maintenance

The final phase focuses on maintaining the deployed system to ensure its continued reliability and usability. Maintenance activities include correcting defects discovered after deployment, improving system performance, and ensuring compatibility with changes in the operating environment. Any new features, functional enhancements, or major changes requested after the completion of the project will be considered as part of subsequent versions of the BioSphere system, preserving the integrity of the approved implementation while allowing future expansion.

4.4 Development Approach

The Bottom-Up Approach is a software development and design strategy in which individual components or modules are developed, tested, and validated independently before being integrated to form larger subsystems and, ultimately, the complete system (Pressman et al., 2020). This approach emphasizes building from the lowest-level functional units toward higher-level functionalities, allowing foundational components to be verified before system-wide integration.

In the development of BioSphere, the Bottom-Up Approach guides the implementation of core modules, such as authentication, and inventory management, which are progressively integrated to construct the complete museum management system. The following figures present selected major modules and do not represent every supporting service or cross-cutting system capability.

As shown below in Figure 19, the Specimen Inventory Management module serves as the core module of the Museum Curator Platform, enabling curators to manage the museum's specimen records. It allows users to create, view, update, archive, and organize specimen information, including taxonomic classifications, specimen categories, storage locations, habitats, conservation status, and specimen images. As the central repository of specimen data, this module provides the information required by other system modules, such as QR Code Management, Reports Generation, and the Augmented Reality system.

Figure 19
Bottom-Up Approach of Specimen Inventory Management Module

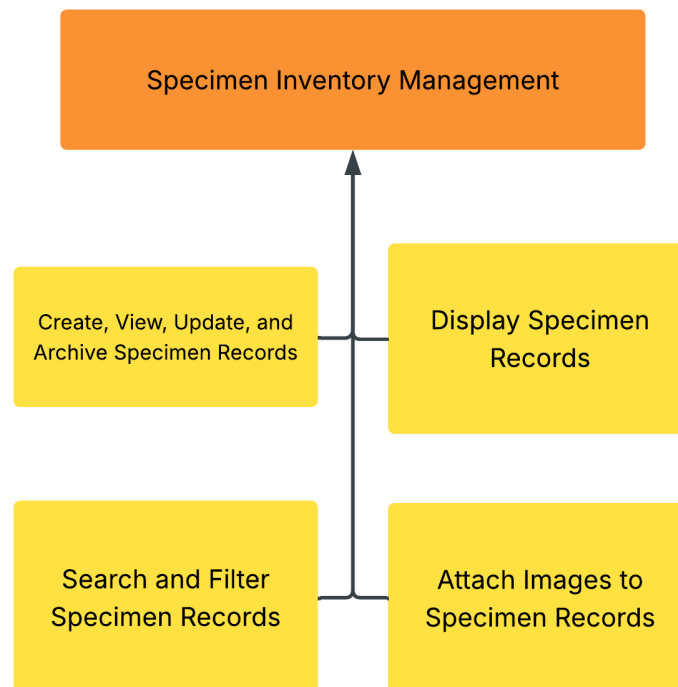
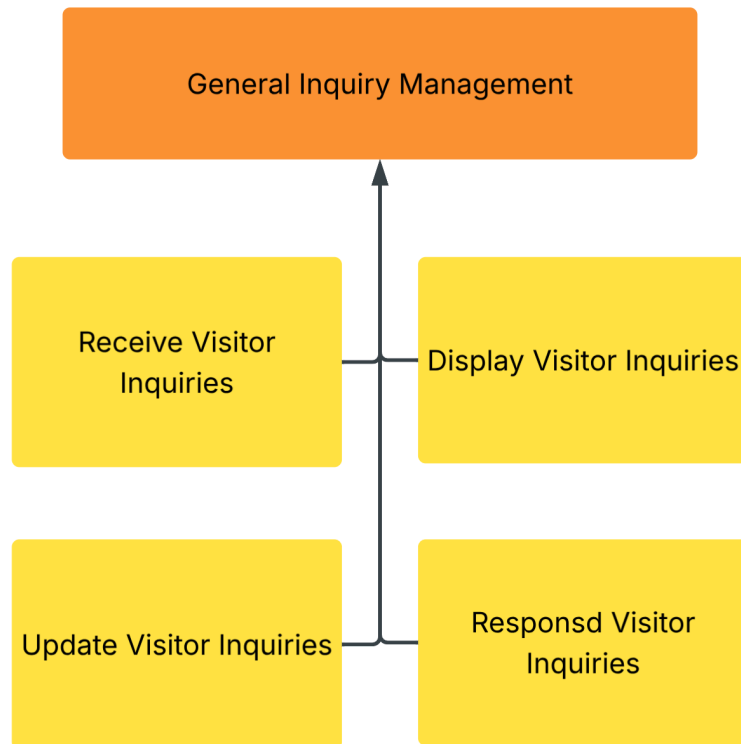


Figure 20

Bottom-up Approach of General Inquiry Management



As shown in figure 20 above, the General Inquiry Management module enables museum curators to review, update, and manage inquiries submitted through the public-facing website. Curators may compose outbound replies through BioSphere, which sends the messages to the visitor's email address and records the available delivery result. Replies received through an external museum email account are not automatically imported and may be manually reflected in BioSphere when necessary.

As shown in figure 21 below, the Museum Visit Request Management module enables curators to manage museum visit requests submitted by visitors from the public facing website. Through this module, curators can review these requests, review each request, approve or decline it, update its status, and track its progress. The component sends outbound email messages and status

updates initiated by the curator through BioSphere. When additional information is required, the visitor may respond through the museum's official email account outside BioSphere, after which the curator may manually update the Visit Request. The system does not automatically retrieve or synchronize incoming email replies.

Figure 21
Bottom-Up Approach of Museum Visit Request Management Module

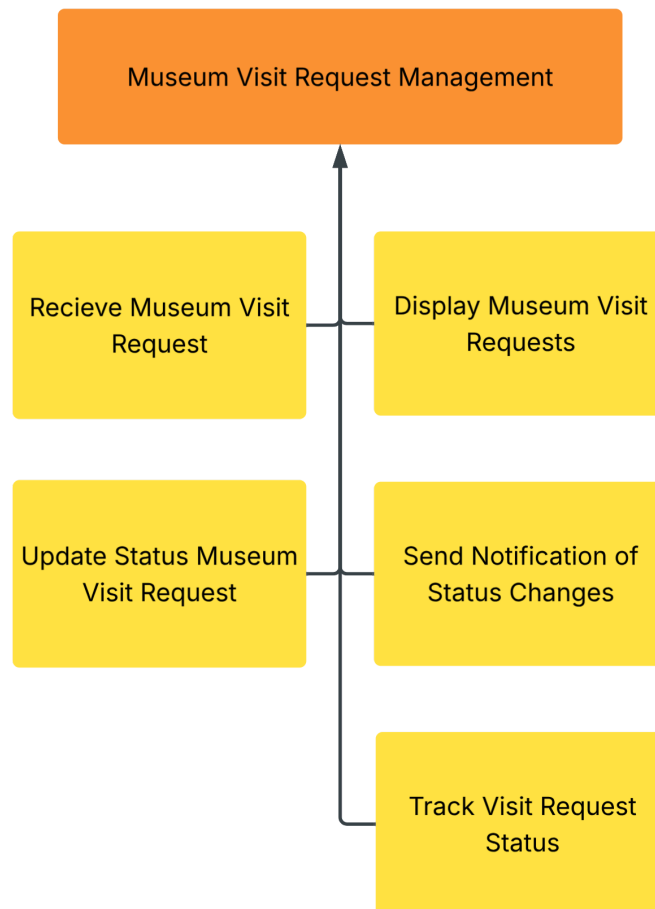
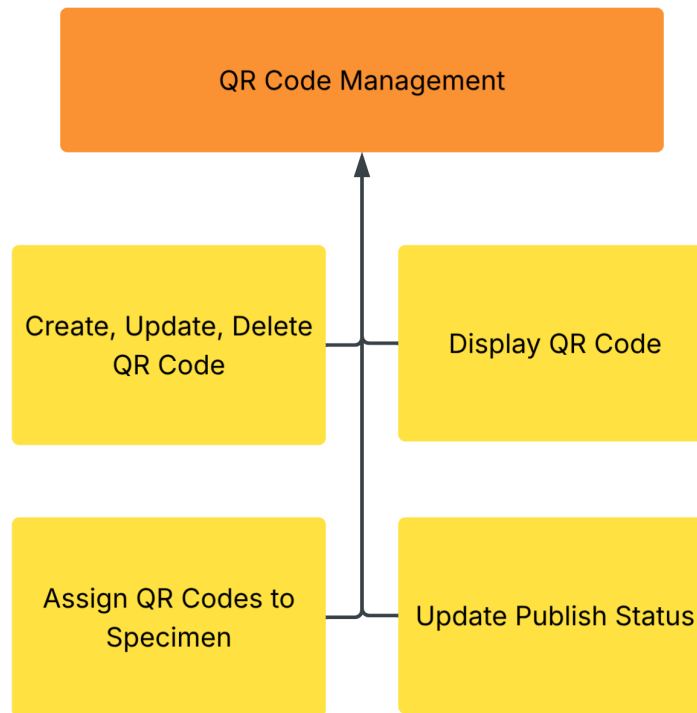


Figure 22
Bottom-up Approach of QR Code Management Module

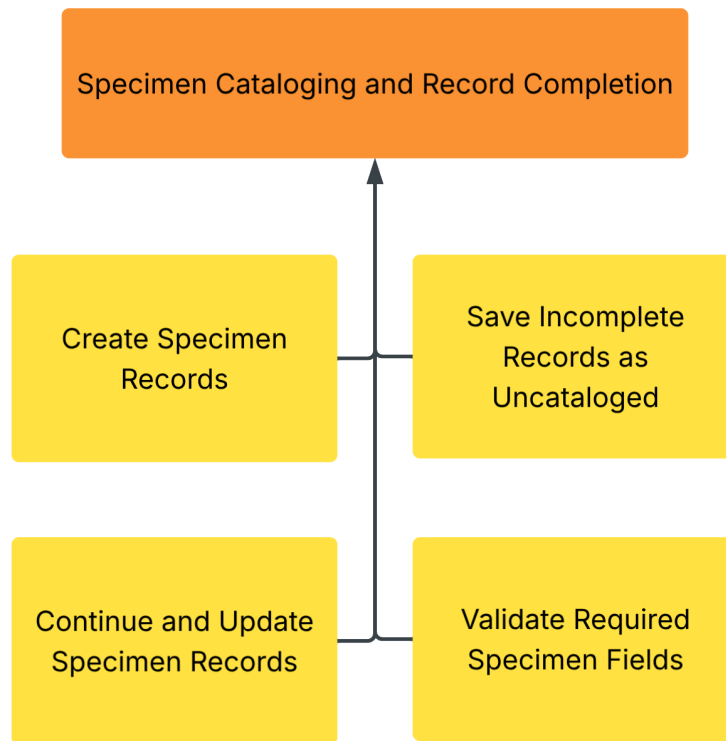


As shown in figure 22, the QR Code Management module enables curators to generate and manage QR codes associated with museum specimens. Each generated QR code is linked to a specimen record, allowing visitors to conveniently access specimen information and augmented reality content.

As shown below in Figure 23, the Specimen Cataloging and Record Completion module enables museum curators to create, save, and update specimen records. Records with missing applicable required information are saved as Uncataloged and may be completed later. The module validates required specimen fields and changes a record to Cataloged once all applicable requirements are completed and verified.

Figure 23

Bottom-up Approach of Specimen Cataloging and Record Completion



As shown below in Figure 24, the Reports and Data Export module enables museum curators to generate inventory reports and summaries of general inquiries and museum visit requests using data stored in BioSphere. The module also allows supported reports to be filtered, printed, and exported in approved file formats for documentation, analysis, and administrative purposes.

Figure 24

Bottom-up Approach of Reports and Data Export

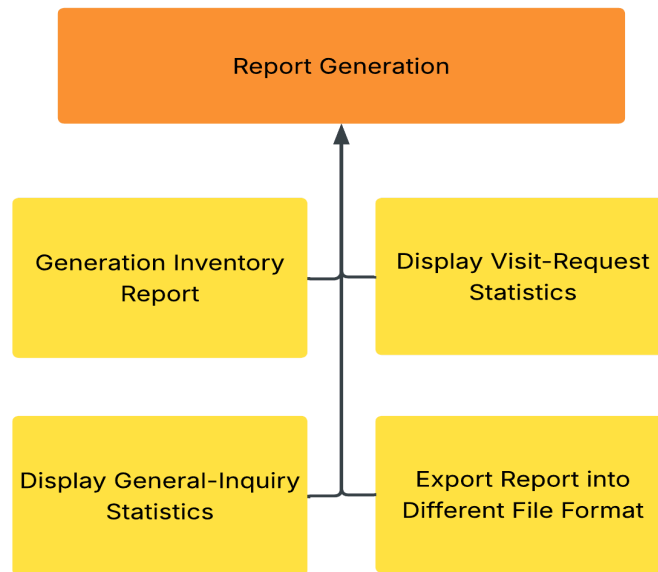
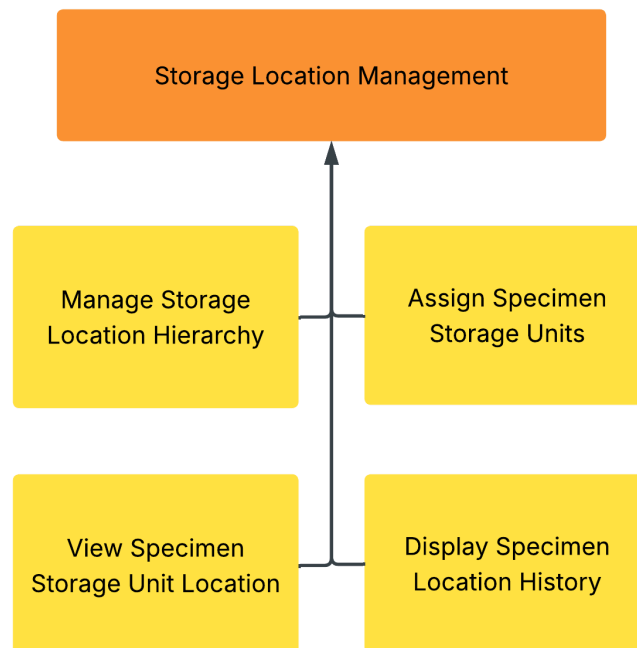


Figure 25

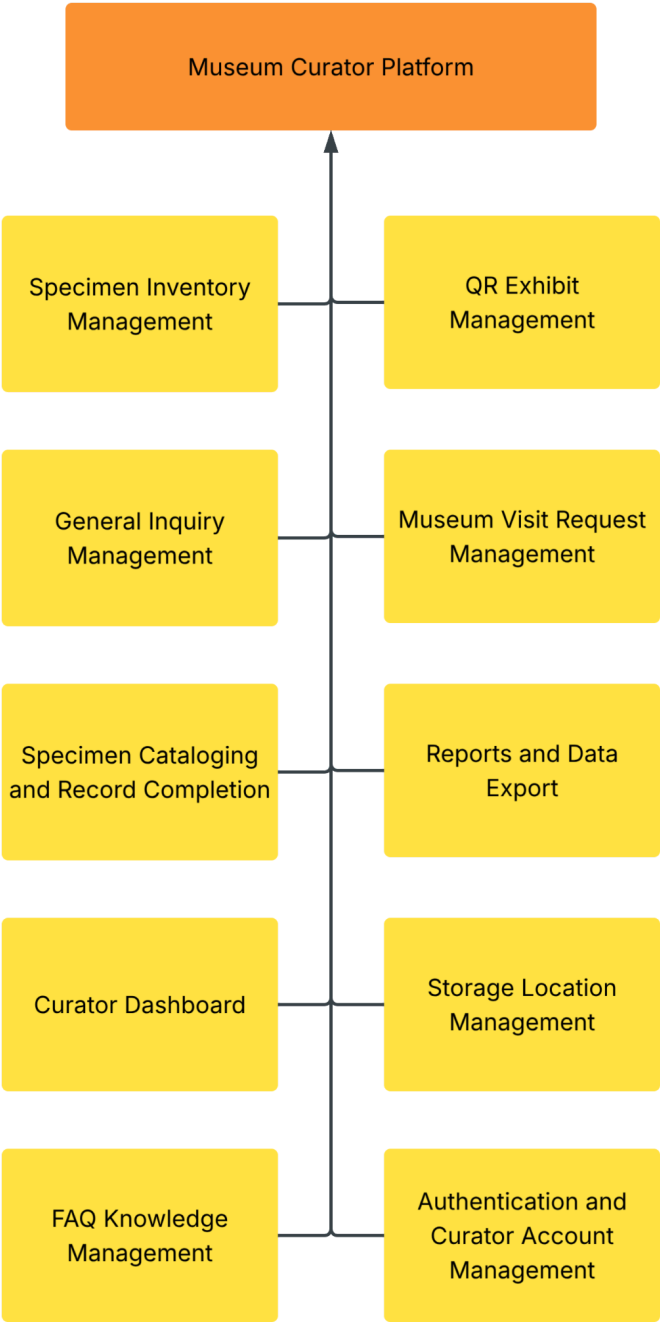
Bottom-up Approach of Storage Location Management



As shown in figure 25, the Storage Location Management module enables museum curators to organize and manage the physical storage locations of museum specimens. It allows the creation and maintenance of storage location hierarchies, assignment of specimens to specific storage units, viewing of specimen storage locations, and tracking of specimen location history to ensure accurate inventory management and traceability.

As shown in figure 26, the Museum Curator Platform integrates the Authentication and Curator Account Management, Curator Dashboard, Specimen Inventory Management, Specimen Cataloging and Record Completion, Storage Location Management, General Inquiry Management, Museum Visit Request Management, QR Exhibit Management, FAQ Knowledge Management, and Reports and Data Export modules into a centralized workspace for authorized curators. Through this platform, curators can manage specimen records and storage locations, process inquiries and visit requests, maintain approved FAQ and QR exhibit content, and generate reports for museum operations.

Figure 26
Bottom-up Approach of Museum Curator Platform

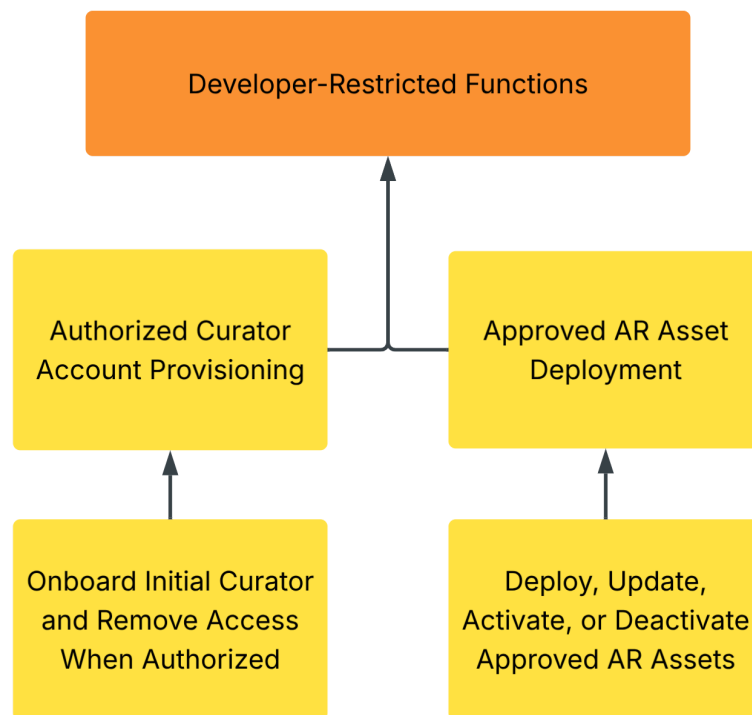


As shown below in figure 27, the Developer-Restricted Functions support only the technical responsibilities assigned to the developer in BioSphere. These include onboarding the initial curator, disabling or removing curator access when

formally authorized, and deploying or updating approved AR assets for selected exhibits. These functions do not provide access to routine museum operations such as specimen management, storage-location management, inquiries, Visit Requests, reports, FAQ-knowledge management, or QR-exhibit content management. Static Home, Gallery, Visit, and About page changes are handled separately through authorized frontend maintenance and redeployment.

Figure 27

Bottom-up Approach of Developer-Restricted Functions



As shown in Figure 28, the Bottom-Up Approach of the BioSphere Visitor Platform provides public visitors with access to approved museum information through the Home, Gallery, Visit, and About pages. Visitors may use the Controlled Knowledge-Based FAQ Assistant to submit museum-related questions and receive responses based only on curator-approved knowledge. When no relevant approved answer is available, the assistant informs the visitor and directs them to the General Inquiry form. Visitors may also submit general

inquiries and museum visit requests, scan QR codes attached to selected exhibits, view approved QR exhibit information, and access WebAR experiences for selected specimens with available AR assets.

Figure 28

Bottom-up Approach of Visitor Platform

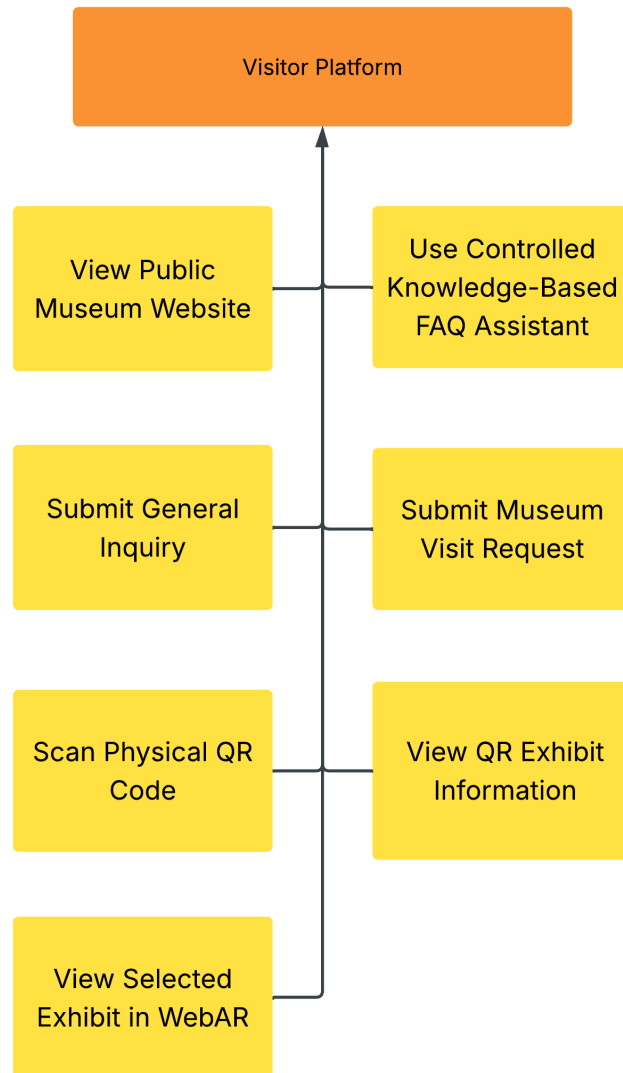
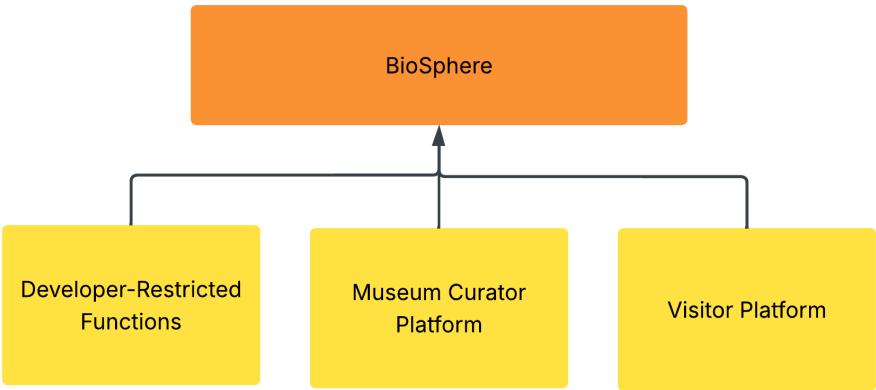


Figure 29
Bottom-up Approach of BioSphere System



As shown in figure 29, the BioSphere System represents the complete integration of the Developer Platform, Museum Curator Platform, and Visitor Platform. By combining these independently developed platforms, the system delivers a comprehensive museum management solution that supports administrative operations, collection management, visitor services, and digital learning through augmented reality. The integration of these platforms demonstrates the culmination of the bottom-up development approach, where individual functional modules are first developed, validated, and then assembled into the final operational system.

4.5 Software Development Tools

As shown, table 3 shows the tools to be used by the proponents to develop BioSphere, along with their version, source, and uses.

Table 3
List of Softwares

Software	Version	Source	Use
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Visual Studio Code	1.129	https://code.visualstudio.com/	Integrated Development Environment
Jira	Cloud	https://www.atlassian.com/software/jira	Planning & Project Management Tool
Figma	Cloud	https://www.figma.com/	Design & Prototyping Tool
GitHub	Cloud	https://github.com/	Version Control
Git	2.50	https://git-scm.com/	Version Control
Lucidchart	Cloud	https://www.lucidchart.com/	Design & Prototyping Tool
SendGrid	Cloud	https://sendgrid.com/	Backend Development
Next.js	16.2.6	https://nextjs.org/	Frontend Development
Tailwind CSS	4.3.3	https://tailwindcss.com/	Frontend Development
Postman	12.9.3	https://www.postman.com/	API

Swagger (OpenAPI)	OpenAPI Spec 3.1	https://swagger.io/	API (OpenAPI)
shadcn/ui	Latest	https://ui.shadcn.com/	Frontend Development
NestJS	11.1.28	https://nestjs.com/	Backend Development
TypeScript	5.9	https://www.typescriptlang.org/	Programming Language (typed JS superset)
Workbox	7.4.1	https://developer.chrome.com/docs/workbox/	Offline-First Web Application (PWA)
Service Worker API	W3C Living Standard	https://developer.mozilla.org/en-US/docs/Web/API/Service_Worker_API	Offline-First Web Application (PWA)
IndexedDB	W3C Living Standard	https://developer.mozilla.org/en-US/docs/Web/API/IndexedDB_API	Database Management
PostgreSQL	18.4	https://www.postgresql.org/	Database Management (RDBMS)
Prisma	7.9.0	https://www.prisma.io/	Object-Relational Mapping
Supabase Storage	Cloud	https://supabase.com/storage	Cloud Object/File Storage Service

Docker	29.6.2 (Engine)	https://www.docker.com/	Containerization and Deployment
Qase	Cloud	https://www.qase.io/	Quality Assurance and Test Management
TRIPO 3D	Cloud	https://www.tripo3d.ai/	AI-Powered 3D Model Generation
Meshy AI	Cloud	https://www.meshy.ai/	AI-Powered 3D Asset Generation (Text/Image-to-3D)

4.6 Project Management

This section presents the project management plan for the development of BioSphere. It covers the proposed schedule and timeline based on the Waterfall Development Model, the distribution of responsibilities among development team members, and the estimated budget required to complete the project.

4.6.1 Schedule and Timeline

Table 4

Academic Year 2026 - 2027, Summer Term

Month	June			July				Aug		
Week	2	3	4	1	2	3	4	1	2	3
Activities										
Requirement Analysis										
Problem Identification										
Requirement Gathering (SRS)										
Literature and Systems Review										

System Design										
DB Schemas, System Architecture and Technical Background										
UI/UX Concept and Prototyping										
Review (Proposal Refinement and Defense)										
Capstone Proposal Defense										
Proposal Refinement and Compliance										

Table 4 presents the project schedule for Academic Year 2026–2027, Summer Semester. During this period, the project focuses on the initial phases of the Waterfall Development Model, specifically Requirement Analysis and System Design, which include activities such as requirements gathering, Software Requirements Specification (SRS) preparation, literature review, database and system architecture design, and UI/UX prototyping. The Review phase is included solely to satisfy the academic requirements of the capstone project, covering proposal refinement, documentation revisions, and the proposal defense. This review phase is not part of the Waterfall software development process and does not affect the sequence of the development phases beyond ensuring that documentation and project milestones are completed within the prescribed academic schedule.

Table 5*Academic Year 2026-2027, First Semester*

Month	Aug			Sep				Oct				Nov	
Week	2	3	4	1	2	3	4	1	2	3	4	1	2
Activities													
Implementation													
Frontend Development													
Backend Development													
Testing													
Functional Testing													
Security Testing													
System-Specific Testing													
Deployment													
User Acceptance													
Final System Review													
Technical Defense													
Maintenance													
Corrective Maintenance													

Table 5 presents the proposed project schedule for Academic Year 2026–2027, First Semester. During this period, the project covers the remaining phases of the Waterfall Development Model, including Implementation, Testing, Deployment, and Maintenance. Implementation focuses on the development of the frontend and backend components, followed by functional, security, and system-specific testing to verify system quality. Deployment activities include user acceptance, final system review, and the capstone technical defense, while the Maintenance phase is limited to corrective maintenance to address issues identified during the evaluation period.

4.6.2 Responsibilities

During the system development phase, tasks are distributed among the group members according to their individual skills, expertise, and availability. This approach aims to improve efficiency, promote teamwork, and allow members to support one another, particularly when addressing complex or technically demanding components of the project.

Table 6

Development Team Responsibilities (DTR)

Name	Role	Assigned Module
Charles Benedict G. Boquecosa	Full Stack Developer	• Curator Dashboard • Visit Request Management • General Inquiry Management • Public Museum Website • BioSphere Backend Management
John Mark B. Gerozaga	Full Stack Developer	• Curator Dashboard • Specimen Inventory Management • Reports and Data Export • General Inquiry Management • Public Museum Website
Kyer Ivan C. Jayme	Full Stack Developer	• Cataloging Management • Storage Location management • Controlled Knowledge-Based FAQ Assistant • Audit Log and Backup/Restore management • Limited Offline Record Caching, Draft Synchronization, and System Integration

		Controlled Knowledge-Based FAQ Assistant
Voltaire Raul L. Ratilla	Full Stack Developer	- Authentication and Access Control - Exhibit and QR Content Management - Public Museum Website - QR Exhibit and WebAR - BioSphere Backend Management

Table 6 presents the distribution of responsibilities among the members of the development team for the implementation of BioSphere. The table identifies the functional modules assigned to each developer and specifies their corresponding technical responsibilities in terms of frontend development (FE), backend development (BE), and deployment (DP). While individual developers are responsible for implementing their designated modules, project management and quality assurance activities are coordinated by all the developers across all modules to ensure effective collaboration, consistent development practices, and adherence to the project's quality standards throughout the software development lifecycle.

To promote effective collaboration and accountability throughout the development process, the project utilizes Jira, GitHub, and Qase as its primary project management, version control, and test management tools. Jira is used for task planning, assignment, progress monitoring, and contribution tracking, allowing the development team to organize project activities and monitor the completion of assigned responsibilities. GitHub serves as the version control platform for source code management and collaborative software development, maintaining a history of code commits, branches, pull requests, and repository changes. Qase is utilized

as the project's test management platform for designing, organizing, and documenting test cases, managing test suites, and recording test execution results to support systematic software testing and quality assurance. Together, these tools provide a transparent mechanism for tracking individual contributions, coordinating development activities, managing testing activities, and ensuring accountability throughout the implementation of BioSphere.

4.6.3 Budget and Cost Management

Table 7
Proposed Budget

Description	Cost
Laptop (4 units)	₱199,981.89
Smartphone (4 units)	₱67,999
Internet Connection (4 months)	₱12,695.00
Printing & Documentation	₱5,000.00
Transportation (4 months)	₱22,000
TRIPO 3D & Meshy AI Subscription (4 months)	₱6,404.00
Contingency	₱20,000.00
Total	₱334,079.89

As shown, table 7 presents the estimated expenses required for the development and implementation of BioSphere. It includes projected costs for project materials, documentation, testing, deployment, and other necessary activities. Existing resources, such as personal laptops, internet

connectivity, and software available through free or educational licenses, are utilized by the development team and do not incur additional project costs.

4.7 Verification, Validation, and Testing

To determine whether BioSphere satisfies its approved requirements and operates as intended, verification, validation, and testing activities will be conducted during and after the implementation phase of the Waterfall Development Model. These activities will evaluate the system's functionality, reliability, security, usability, and readiness for pilot deployment. The testing process will include unit testing, integration testing, black-box testing, security testing, system-specific testing, and User Acceptance Testing.

Email testing will verify curator alerts, curator-initiated outbound inquiry replies, Visit Request messages and status updates, delivery-result handling, and the absence of external mailbox access or automatic incoming-email synchronization.

Verification will determine whether BioSphere has been developed according to the functional and technical requirements stated in the approved Software Requirements Specification. Unit testing will be performed by the system developers as individual components are completed to verify that each module, function, and validation rule operates correctly. Integration testing will then be conducted to confirm that interconnected modules communicate and exchange data properly without functional conflicts.

Black-box testing will serve as the primary functional verification method and will be conducted by four (4) developers. The system will be evaluated using valid, invalid, and boundary inputs, and the actual results will be compared with the expected results defined in the approved requirements. This approach will assess the system from the user's perspective without examining its internal

source code and will help identify incorrect outputs, missing functions, validation errors, and unexpected system behavior.

System-specific testing will evaluate the limited offline functionality of BioSphere. It will verify read-only access to previously synchronized specimen records, local searching and filtering, creation and preservation of new Uncataloged text-only drafts, successful draft synchronization after reconnection, retention of drafts following failed synchronization, prevention of duplicate records during synchronization retries, and blocking of unsupported offline editing operations.

Validation will be conducted through User Acceptance Testing to determine whether BioSphere supports the operational needs and expectations of its intended users. Unlike verification, which determines whether the system was implemented according to its specifications, validation will assess whether the completed system is appropriate and usable for actual museum operations.

The museum curators will evaluate the authenticated application, including specimen cataloging, inventory management, storage-location management, visit requests, general inquiries, reports, curator account management, and QR exhibit management. Two (2) curators will serve as UAT participants for the specimen records management system since they are the system's primary intended users. As for the public-facing website, approximately fifty (50) participants will be involved in UAT, a figure based on the average frequency of physical visitors to the museum as confirmed by the curator. Available museum visitors or representative public users will evaluate the public website, Controlled Knowledge-Based FAQ Assistant, inquiry and visit-request forms, QR-linked exhibit pages, and selected AR experiences. The system developers will conduct and document the technical testing activities but will not replace the intended users during UAT.

Participants will evaluate the applicable system components based on functionality, usability, responsiveness, ease of navigation, clarity of information, and overall user experience. Test results, identified defects, and participant feedback will be documented and analyzed. Critical defects and incorrect implementations of approved requirements will be corrected before pilot deployment, while new feature requests outside the approved scope will be documented for future development.

QR and AR compatibility testing will verify the accessibility of exhibit content across representative supported mobile devices and browsers. Testing will include successful QR access, manual direct-URL access, device and browser compatibility detection, camera-permission handling, supported AR launch, unsupported-device behavior, AR loading failure, and fallback to the regular exhibit page. Representative Android and iOS devices will be included where available, together with an older or lower-capability available device to verify that unsupported AR functionality does not prevent access to the approved specimen information.

Through these complementary testing methods, the researchers aim to establish that BioSphere has been developed according to its approved requirements and can support the operations and public services of the University of San Carlos Biological Museum.

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Appendix A
Transmittal Letter



June 27, 2026

Fidel Bendanillo
Curator
USC Biological Museum,
Dear **Mr. Bendanillo**,

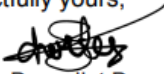
Greetings!

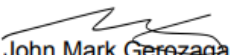
We, the undersigned third-year Bachelor of Science in Information Technology students of the University of San Carlos, respectfully submit our capstone project proposal entitled **"BioSphere: An Offline-First Web-Based Biological Specimen Inventory, Visitor Request, And Augmented Reality Exhibit System For University Of San Carlos Biological Museum"** for your review and consideration.

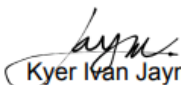
The proposed study aims to develop an offline-first, web-based system for the University of San Carlos Biological Museum, integrating specimen inventory management, visitor request handling, and augmented reality exhibits to improve the organization, tracking, and accessibility of its collections. We respectfully seek your permission and cooperation to conduct this study and gather the information necessary for the successful development of the proposed system.

Thank you for your time and consideration. We look forward to your favorable response.

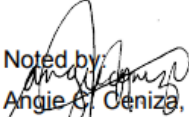
Respectfully yours,

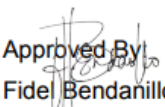

Charles Benedict Boquecosa


John Mark Gerozaga


Kyer Ivan Jayme


Voltaire Raul Ratilla

Noted by:

Angie C. Ceniza, Ph.D.
Capstone Project Adviser

Approved By:

Fidel Bendanillo
Curator

Appendix B

Interview Guidelines

1. Meeting Details

Date: July 17, 2026

Time: 1:30 PM - 4:30 PM

Location: 3rd Level, Learning Resource Center, Josef Baumgartner Building, USC, Nasipit, Talamban Campus, Cebu City, Philippines, 6000

Minute Taker: Voltaire Raul Ratilla

Attendees: Charles Benedict Boquecosa, John Mark Gerozaga, Kyer Ivan Jayme

2. Agenda

- Introduction to the University of San Carlos Biological Museum
- Discussion of the museum's current specimen-recording process
- Identification of problems in the current inventory and backup process
- Discussion of specimen storage and location management
- Discussion of visitor inquiries and visit requests
- Identification of requirements for the proposed BioSphere system
- Discussion of the public website, FAQ Assistant, QR exhibits, and Augmented Reality
- Discussion of reports, notifications, offline access, security, and system maintenance

3. Main Discussion Areas

Current Specimen-Recording Process

The team discussed how specimen information is currently recorded using physical logbooks and spreadsheet files. The curator's responsibilities and the participation of supervised OJT students in encoding specimen information were also reviewed.

Current Problems and Limitations

The team identified the lack of a reliable backup mechanism for locally stored spreadsheet files as one of the main concerns. Incomplete specimen information, manual record management, and the possible loss or corruption of locally stored records were also discussed.

Specimen Cataloging and Identification

The discussion covered the required specimen information, accession-number format, taxonomic classification, specimen categories, condition categories, quantity grouping, and the handling of incomplete specimen records.

Storage and Location Management

The team discussed how specimens are physically arranged and labeled within rooms, cabinets, drawers, shelves, boxes, containers, and display areas. The proposed system's location hierarchy and specimen-movement functions were also reviewed.

Visitor Inquiries and Visit Requests

The current process for receiving inquiries and visit requests through email, telephone, social media, personal messages, walk-in requests, and university communication channels was discussed. The curator also explained the approval, scheduling, notification, and campus-entry processes.

Public Website and FAQ Assistant

The team discussed the information that should be available on the public museum website, including museum details, operating hours, contact information, visitor guidelines, and gallery content. The proposed FAQ Assistant was also discussed as a feature that would answer questions using curator-approved museum information.

QR-Based Exhibits and Augmented Reality

The discussion covered the selection of specimens for QR exhibit pages, the information that may be displayed publicly, QR-code generation, printing, and access. Selected specimens for Augmented Reality presentation and the expected fallback when AR is unavailable were also reviewed.

Reports, Notifications, and Offline Access

The team discussed the required inventory, visitor, inquiry, audit, quarterly, and annual reports. Email, SMS, and in-system notifications were considered during the meeting. After subsequent scope review, SMS integration was removed from the BioSphere MVP, while email and in-system notifications were retained together with offline specimen access, synchronization, backup, and recovery.

Users, Security, and Maintenance

The responsibilities and permissions of the Museum Curator and Developer were discussed. The team also considered account management, access restrictions, audit logging, system turnover, ownership, and maintenance after the completion of the project.

4. Brief Proceedings

Discussion of the Current Museum Process

The team reviewed the existing procedures for recording, classifying, storing, locating, and retrieving specimen information. The use of physical logbooks, spreadsheets, labels, accession numbers, and storage arrangements was discussed.

Identification of System Requirements

The curator's needs and expectations for the proposed BioSphere system were reviewed. The discussion focused on specimen cataloging, storage-location

management, backup and recovery, visitor coordination, inquiries, reporting, and public information.

Discussion of Visitor-Facing Features

The team discussed the proposed public museum website, FAQ Assistant, QR-based exhibit pages, and selected Augmented Reality content. The information that may be shown publicly and the curator's role in approving museum content were also considered.

System Proposal Discussion

The proposed BioSphere system was presented as an offline-first biological museum management system designed to support specimen management, visitor services, reporting, QR-based exhibits, selected AR content, notifications, audit logging, and backup and recovery.

Clarification of Pending Requirements

Questions that were not fully answered during the interview were identified for further clarification and validation with the Museum Curator before the finalization of the Software Requirements Specification.

Appendix C
Interview Transcript July 8, 2026

Project Title	BioSphere: An Offline-First Biological Museum Management System with QR-Based Exhibits and Augmented Reality
Interviewee	Mr. Fidel Bendanillo
Position	Curator, USC Biological Museum
Meeting Date	July 8, 2026
Day and Time	Wednesday, 2:00 PM
Meeting Type	Formal Interview

Preliminary Concerns Raised by the Curator

- Who will take responsibility for maintaining the system after the original development team graduates?
- Where will the budget for system maintenance, hosting, and future improvements come from?
- How will the system accommodate the curator, considering that he is not highly familiar with technology?

These concerns indicate the need for a simple and user-friendly interface, proper system turnover, user orientation, maintenance documentation, and a clear deployment and sustainability plan.

Interview Questions and Responses

1. What is the current process and structure for creating a specimen entry or record?

Response: Specimen information is currently recorded in two formats:

- Written manually on paper
- Encoded digitally in a Microsoft Excel spreadsheet
- The specimen entry may include taxonomic information, classification, donor information, accession information, and preservation status.
- For accessioning, newly acquired specimens receive an accession code. When another specimen of the same species is received, the existing accession code may be used and the recorded specimen quantity is increased.

2. Approximately how many specimens are currently displayed in the Main Gallery?

Response:

- Stuffed vertebrate animals: approximately 110
- Insects: more than 1,000
- Corals: approximately 500

3. How frequently does the museum receive new specimens?

Response: The museum currently receives fewer than 100 specimens because specimen acquisition and collection are affected by government policies and regulatory requirements.

4. What are the usual sources of the museum's specimens?

Response:

- Students
- Individuals working in the same field or sharing similar interests
- Private individuals
- Most donated or collected specimens are species commonly encountered in the wild.

5. Does the museum conduct field expeditions?

Response: Yes. The museum conducts field expeditions in different locations, including areas outside Cebu and in other parts of the Philippines. However, these activities must comply with applicable local government requirements, permits, and regulations.

6. What does the curator do when errors are found in a specimen entry?

Response:

- When specimen information is uncertain or incorrect, the curator consults taxonomy references available online, including taxonomy guides and Wikipedia, to verify the specimen details.

- For written specimen records, the genus and species names are underlined, and incorrect information is corrected while preserving the previous entry when necessary.
- For digital specimen records, the genus and species names are italicized and incorrect details are updated in the digital record.
- For collector labels, the old label is not removed. A new or corrected label is placed above or below the original label to preserve the record of the previous information.

7. How does the curator currently locate a specimen when it is needed?

Response: The curator relies mainly on his familiarity with and memory of the specimens' physical locations within the museum.

8. What challenges are encountered in the current specimen-management process?

Response: No complete response was recorded for this question during the interview. Additional validation with the curator is required.

Key Information Requiring Further Verification

- The official accession-number format
- Whether specimens of the same species always share an accession code
- The exact rules for increasing specimen quantity
- The required fields for a complete specimen record
- The specific preservation-status values used by the museum
- The curator's main challenges in managing paper and spreadsheet records
- The preferred system-maintenance and turnover arrangement
- The approved budget and hosting responsibility after project completion

9. Approximately how many and how frequent do visitors attend the main gallery?

Response: Visitors attending educational tours commonly arrive in groups of approximately fifty (50) to one hundred (100) visitors. The frequency of these tours depends on scheduled museum visits and educational activities.

Appendix D
Condensed Interview Transcript

Note: This document is a condensed interview transcript based on the meeting notes and recorded responses gathered during the interview conducted on July 17, 2026. It summarizes the main questions, answers, and discussion points and does not represent the exact word-for-word statements of the participants. Additionally, this document does not reproduce the participants' exact words. Requirements mentioned during the meeting were subject to subsequent validation and scope revision.

1. Meeting Details

Date: July 17, 2026

Time: 1:30 PM–4:30 PM

Location: 3rd Level, Learning Resource Center, Josef Baumgartner Building, University of San Carlos, Nasipit, Talamban Campus, Cebu City, Philippines 6000

Minute Taker: Voltaire Raul Ratilla & Ivan

Attendees:

- Charles Benedict Boquecosa
- John Mark Gerozaga
- Kyer Ivan Jayme
- Museum Curator

2. Interview Transcript

Question 1: How are specimen records currently created, maintained, classified, and located in the museum?

Curator: Specimen information is initially written in physical notebooks or logbooks and is later encoded into Excel or spreadsheet files. The curator is mainly responsible for recording and updating specimen information. OJT students may assist with encoding, but their work must be supervised and approved by the curator.

Specimens are classified according to their taxonomy. The museum records information such as the accession number, common name, scientific name, collection date, collection location, collector, acquisition source, preservation method, condition, storage location, images, and remarks. Full taxonomic classification, from kingdom to species, is also used.

The official accession-number format and assignment procedure remain subject to curator validation. Existing specimen numbers should be retained. Specimens belonging to the same group or species may share an accession number, while their quantity is recorded under the same entry.

The physical location of specimens is determined through labels placed on cabinets, drawers, containers, and specimen records. These labels may include the cabinet number, drawer number, taxonomic information, and accession number. The curator may also refer to the existing Excel records when locating specimens.

Question 2: What problems or limitations are experienced in the current inventory, backup, and storage processes?

Curator: The primary concern is the absence of a dependable backup for spreadsheet files stored locally on a computer. If the device experiences a hardware problem or if a file becomes corrupted, the specimen records may no longer be available. The proposed system should provide a way to back up and recover these records.

Some specimens are received with incomplete information or without proper labels. However, a specimen may still be recorded when the basic required information is available. The curator will determine whether an incomplete entry has enough information to be treated as an official specimen record.

Searching for specimens is not currently considered a major problem because the curator is familiar with the collection and specimens are already organized by

taxonomy and physical location. However, relying on the curator's memory may become difficult as the collection grows or when another person assumes responsibility for the museum.

The physical storage arrangement includes areas and units such as the receiving area, main gallery, herbarium area, wet area, cabinets, drawers, shelves, boxes, jars, display cases, and other containers. Each storage unit should have a readable and unique label. The proposed system should display the complete location path and the number of specimens stored in each unit.

The curator should be able to move an individual specimen or an entire storage unit. When a cabinet or another parent storage unit is moved, all storage units and specimens contained within it should be treated as part of the new location. A storage unit containing specimens should not be permanently deleted until the specimens have been transferred or the location has been properly archived.

Question 3: How are visitor inquiries and museum visit requests currently received and processed?

Curator: Visitors currently contact the museum through university email, telephone, the curator's cellphone, Facebook or other social media, personal messages, walk-in requests, or communication forwarded through the university secretary.

Common visitor questions include the museum's history, office hours, preservation methods, information about the curator, whether displayed specimens are genuine, and the history of particular species or specimens. Large tour groups may also experience difficulty hearing verbal explanations clearly.

Visit requests currently require communication between the visitor, curator, and university personnel. This can result in unnecessary back-and-forth communication. The curator normally approves, rejects, or reschedules requests

through email or telephone. The final schedule and campus-entry information are then sent to the requester.

The proposed system should collect the required requester and visitor information, including the preferred date and time, group or school name, visitor names, and number of visitors. A requester should be allowed to submit up to three preferred schedules, with one required schedule and two optional schedules.

The system should prevent scheduling conflicts, although one visit in the morning and another in the afternoon may be allowed. The curator may also propose a different schedule. Approved, completed, cancelled, or rescheduled visits should remain in the system for reporting.

BioSphere should generate a printable or exportable summary that the curator can use when manually submitting campus-entry information to USC or ISMIS. The system will not directly integrate with or replace the university's campus-entry system.

Question 4: What public website, FAQ Assistant, QR exhibit, and Augmented Reality features are required?

Curator: The public museum website should introduce the museum and provide information such as its history, mission and vision, curator information, office hours, contact details, location, visitor guidelines, gallery content, and procedures for inquiries and visit requests.

The public website should not provide a searchable catalog of the museum's complete internal specimen inventory. Internal storage locations, accession numbers, condition notes, private donor information, requester information, account information, and other sensitive records must not be displayed publicly.

The FAQ Assistant should answer frequently asked museum-related questions using information approved by the curator. These questions may include the

museum's history, operating hours, preservation methods, collection sources, and other basic visitor concerns. When the FAQ Assistant cannot find an approved answer, it should direct the visitor to the General Inquiry form. It should not approve or process visit requests.

QR exhibit pages should be created only for selected specimens chosen by the curator. The QR page may show approved information such as the common name, scientific name, description, habitat, ecological role, conservation status, fun facts, images, videos, and general collection information. Each selected exhibit should have one QR code that the curator can download and print.

QR exhibit pages may be saved, published, unpublished, edited, and republished. The same QR code should continue to direct visitors to the exhibit page after its content is updated. QR exhibit pages should be accessible through their QR code or direct link rather than through a public searchable specimen catalog.

Augmented Reality should be limited to selected approved specimens. Possible initial specimens discussed included the python, six-legged carabao, pangolin, whale, fox, opossum, civet cat, and monkey. The AR experience should be accessible through a button on the QR exhibit page. When the visitor's device does not support AR, the normal QR exhibit information should remain available.

Question 5: What requirements were identified for reports, notifications, offline access, security, and system maintenance?

Curator: The system should generate reports such as the complete specimen inventory, specimens by category, specimens by location, specimens by condition, incomplete records, recently added specimens, visitor summaries, inquiry summaries, quarterly reports, annual reports, audit logs, QR scan counts, number of collections, number of classified species, number of donated specimens, and number of visitors.

Reports should support suitable filters and may be exported as PDF, CSV, Excel, or printable pages. Only the curator should be authorized to generate and export museum reports.

The curator should receive notifications for new visit requests, upcoming visits, new inquiries, incomplete records, and backup problems. During the discussion, in-system, email, and possible SMS notifications were considered for notifying the curator about new inquiries and visit requests. After subsequent scope review, the project team removed SMS integration from the BioSphere MVP. The approved notification channels are in-system alerts and email notifications.

The internal specimen-management system should remain usable during temporary internet interruptions. Important offline functions may include viewing cached specimen records, searching records, editing specimen information, viewing storage locations, and recording changes. Public website access, QR pages, email notifications, cloud backup, incoming requests, and online AR assets may require an internet connection.

When the internet connection returns, offline changes should be synchronized. The curator should be informed when records are waiting to synchronize or when synchronization fails.

The Museum Curator will manage specimen information, storage locations, QR exhibit content, inquiries, visit requests, and reports. Developer access should be restricted to authorized account onboarding, curator account succession, system deployment, approved updates, and deployment of approved AR assets. The Developer should not modify specimen information or curator-controlled museum content.

Accounts should be protected through access controls, strong passwords, session security, audit logging, backup and recovery, and account deactivation when access is no longer authorized. The responsibilities for hosting, ownership,

maintenance, system turnover, and curator succession must be documented before deployment.

Subsequent scope clarification: Following feasibility review, the MVP offline scope was limited to viewing and searching previously synchronized specimen records and creating new Uncataloged text-only drafts. Offline editing of existing records, storage-location changes, quantity or condition updates, media uploads, and conflict-resolution functions were deferred beyond the MVP.

3. Closing Summary

The interview provided information about the museum's current specimen-recording, storage, visitor-coordination, and reporting processes. It also identified the required functions of BioSphere, including specimen cataloging, storage-location management, backup and recovery, visitor inquiries, visit requests, the public museum website, the FAQ Assistant, QR exhibit pages, selected Augmented Reality content, reporting, notifications, offline access, security, and account succession.

Some requirements remained subject to additional validation, including the final list and number of AR exhibits, exact public content, visitor-data retention periods, offline synchronization rules, backup procedures, university branding requirements, hosting arrangements, and long-term system maintenance.

Appendix E

Software Specification Requirements

CURRICULUM VITAE

CONTACT INFORMATION

1. Full Name: Kyer Ivan Jayme
2. Address: Nasipit, Talamban, Cebu City
3. CellPhone: +63 0966 573 9589
4. Email: jaymekyerivan@gmail.com



PERSONAL INFORMATION

1. Date of Birth: Aug 4, 2004
2. Age: 21
3. Citizenship/Nationality: Filipino
4. Gender: Male
5. Marital Status: Single
6. Number of Children & Age: N/A
7. Language proficiency (*state verbal and/or written*)
 - a. Verbal and written English - C1
 - b. Verbal and written Filipino - Native / Bilingual Proficiency
 - c. Verbal and written Cebuano - Native / Bilingual Proficiency
8. Computer PMS and Software skills
 - a. **Productivity Tools:** Microsoft Office, Google Workspace, Google Drive, Gmail, and Google Calendar.
 - b. **Project Management and Collaboration:** Trello, Notion, Slack, Microsoft Teams, and Frame.io.
 - c. **Data and Reporting:** Power BI, Microsoft Excel, Google Sheets, data cleaning, reporting, and dashboard preparation.
 - d. **Web Development:** React, JavaScript, HTML, CSS, GitHub, Supabase, MySQL, PostgreSQL, and basic API integration.
 - e. **Cloud and Systems:** AWS S3, Windows, Linux/Ubuntu, VirtualBox, VMware, Whonix, Tor Browser, and basic troubleshooting.
 - f. **Creative Tools:** Adobe Premiere Pro, CapCut, Canva, and basic social media content editing.

EDUCATIONAL BACKGROUND

1. Education Level

2023 - Current

University of San Carlos

Testiary

2021-2023

De La Salle University - Ozamiz

Secondary

2. Certifications & Accreditations

Data Analyst Associate

March 2026

Datacamp

NC II in Computer Systems Servicing

Dec 2024 - Dec 2029

Python and AWS Elective Course

May 2025

Zuitt - Coding Bootcamp

CCNAv7: Switching, Routing, and Wireless Essentials

Jan 2025

Cisco Networking Academy

SQL Associate

Aug 2024-2026

Datacamp

STRENGTHS/TRAITS & SKILLS

1. Hands-on technical experience
2. Strong analytical and problem-solving skills
3. Good communication and interpersonal skills
4. Able to work under pressure and within tight schedules
5. Adaptable and willing to learn new tools
6. Detail-oriented in documentation and reporting
7. Able to work independently and in a team
8. Good time management and task prioritization
9. Professional, responsible, and open to feedback

CAREER OBJECTIVES

To develop a career in the data field, starting as a Data Analyst and progressing toward Data Engineering and other data-related roles. I aim to apply my skills in data analysis, reporting, databases, and technology to support business decisions, improve processes, and contribute to organizational growth.

REFERENCES

1. Kim Christian Jayme
Brother
Accenture
2. Angie M. Ceniza - Canillo
Adviser
University of San Carlos, Talamban Campus

Resume updated on July 22, 2026

CONTACT INFORMATION

1. Full Name: Voltaire Raul L. Ratilla
2. Current Address: La Montaña Homes,
Cabaangahan, Consolacion, Cebu
3. Cellphone number: +63 09912117068
4. Email address: voltaireratilla@gmail.com



PERSONAL INFORMATION

1. Date of Birth: February 11, 2003
2. Age: 23
3. Citizenship/Nationality: Filipino
4. Gender: Male
5. Marital Status: Single
6. Number of Children & Age: N/A
7. Language proficiency
 - Verbal and written English - Native / Bilingual Proficiency
 - Verbal and written Filipino - Native / Bilingual Proficiency
 - Verbal and written Cebuano - Native / Bilingual Proficiency
8. Computer PMS and Software skills
 - Programming languages:
 - C, Java, Python, Javascript, Typescript
 - Frontend Technologies:
 - HTML, CSS, TailwindCSS, Next.js, React.js, AndroidSDK(Java/XML),
 - Backend Technologies:
 - Nest.js, Express.js,
 - Cloud Services and Platforms:
 - Amazon Web Services (AWS), Supabase
 - Database:
 - MySQL, PostgreSQL, MSSQL, SQLITE
 - Operating Systems:
 - Windows, Linux

EDUCATIONAL BACKGROUND (state year achieved)

1. Education Level

2023 - Present

University of San Carlos

Testiary

2. Certifications & Accreditations

May 17, 2025

Zuitt Certification of Completion - Python Fundamentals, Django Fundamentals, AWS EC2, AWS S3, AWS Lambda

Zuitt - Coding Bootcamp

January 22, 2025

CCNAv7: Switching, Routing, and Wireless Essentials

Cisco Networking Academy

STRENGTHS/TRAITS & SKILLS

1. Strong foundation in network infrastructure and cybersecurity fundamentals
2. Hands-on experience with Cisco networking tools and technologies (CCNA-trained)
3. High degree of initiative in self-directed technical learning
4. Analytical and problem-solving mindset for troubleshooting systems

CAREER OBJECTIVES

To become a well-rounded System Administrator with strong, balanced expertise in both network infrastructure and cybersecurity, contributing to the security and efficiency of an organization's IT systems while continuously growing technical skills.

Resume updated on July 27, 2026

CONTACT INFORMATION

1. Full Name: Charles Benedict G. Boquecosa
2. Current Address: Babag 1, Lapu-Lapu City, Cebu
3. Cellphone number: +63 0908-268-5190
4. Email address: charlesbenedictboquecosa@gmail.com



PERSONAL INFORMATION

1. Date of Birth: September 8, 2004
2. Age: 21
3. Citizenship/Nationality: Filipino
4. Gender: Male
5. Marital Status: Single
6. Number of Children & Age: N/A

7. Language Proficiency

- Verbal and written English - Native / Bilingual Proficiency
- Verbal and written Filipino - Native / Bilingual Proficiency
- Verbal and written Cebuano - Native / Bilingual Proficiency

8. Computer PMS and Software skills

- Programming languages:
 - C, C#, Java, Python, Javascript, Typescript
- Frontend Technologies:
 - HTML, CSS, TailwindCSS, Next.js, React.js, AndroidSDK(Java/XML),
- Backend Technologies:
 - ASP.NET Web API, MVC, Nest.js, Express.js,
- Cloud Services and Platforms:
 - Amazon Web Services (AWS), Supabase
- Database:
 - MySQL, PostgreSQL, MSSQL, SQLITE
- Operating Systems:
 - Windows, Linux

EDUCATIONAL BACKGROUND

1. Education Level

2023 - Current

University of San Carlos

Tertiary

2021 - 2023

Saint Alphonsus Catholic School

Secondary

2. Certifications & Accreditations

NC II in Computer Systems Servicing

Dec 2024 - Dec 2029

Python and AWS Elective Course

May 2025

Zuitt - Coding Bootcamp

CCNAv7: Switching, Routing, and Wireless Essentials

Jan 2025

Cisco Networking Academy

Cybersecurity Essentials

Nov 2025

Cisco Networking Academy

STRENGTHS/TRAITS & SKILLS

1. Strong foundation in cybersecurity and network administration
(CCNAv7-certified)

2. Full-stack development experience across Next.js, NestJS, and PostgreSQL
3. Hands-on project leadership — led authentication/access control development for a capstone system
4. Adaptable under pressure — placed 3rd in an international hackathon (Ho Chi Minh City) building an app idea from scratch in a limited timeframe
5. Self-directed learner — independently built a full incident response management system outside of coursework

CAREER OBJECTIVES

To build a career in cybersecurity and network operations, applying my academic training and hands-on project experience toward roles with growing responsibility, with the long-term goal of working internationally in the field.

CONTACT INFORMATION

1. **Full Name:** John Mark Bandalan Gerozaga
2. **Current Address:** 7-4, Main St.,
White Hills Subdivision, Banawa, Brgy.
Guadalupe, Cebu City, Cebu, 6000,
Philippines
3. **Contact Number:** 09614893338
4. **Email Address:** jm.gerozaga02@gmail.com



PERSONAL INFORMATION

1. **Date of Birth:** August 30, 2002
2. **Age:** 23
3. **Citizenship/Nationality:** Filipino
4. **Gender:** Male
5. **Marital Status:** Single
6. **Language proficiency**
 - Verbal and written English - Native / Bilingual Proficiency
 - Verbal and written Filipino - Native / Bilingual Proficiency
 - Verbal and written Cebuano - Native / Bilingual Proficiency
7. **Computer PMS and Software skills**
 - Programming Languages: C, C#, Java, Python, Javascript, Typescript
 - Frontend Development: HTML, CSS, TailwindCSS, Next.js, React.js, AndroidSDK(Java/XML)
 - Backend Development: Express.js
 - Database: MySQL, PostgreSQL, Microsoft SQL Server
 - UI/UX Tools: Figma (basic/intermediate if applicable)
 - Microsoft Office Suite (Word, Excel, PowerPoint)
 - Network Simulation and Virtualization Tools: Cisco Packet Tracer, Oracle VirtualBox, VMware, GNS3

EDUCATIONAL BACKGROUND

1. Education Level

2023 - present

University of San Carlos – Talamban Campus

Bachelor of Science in Information Technology

Tertiary Level

2021 - 2023

Southwestern University – PHINMA

Senior High School

2018 - 2021

Southwestern University – PHINMA

Junior High School

2. Certifications and Accreditations

CCNAv7: Introduction to Networks

Jul 2024

Cisco Networking Academy

NC II in Computer Systems Servicing

Dec 2024 - Dec 2029

Python and AWS Elective Course

May 2025

Zuitt - Coding Bootcamp

CCNAv7: Switching, Routing, and Wireless Essentials

Jan 2025

Cisco Networking Academy

CyberOps Associate

Aug 2025

Cisco Networking Academy

Cybersecurity Essentials
Nov 2025
Cisco Networking Academy

STRENGTHS/TRAITS & SKILLS

1. Effective Time Management and Task Prioritization
2. Quick Learner with Adaptability to New Technologies
3. Strong Programming and Problem-Solving Skills
4. Goal-Oriented with Efficient Workload Management
5. Reliable, Responsible, and Committed to Delivering Quality Work

CAREER OBJECTIVES

Motivated Information Technology student seeking to build a career as a cybersecurity generalist by applying technical skills, continuously expanding cybersecurity expertise, and contributing to the protection of organizational systems and data.

Resume updated on July 27, 2026